

Data Mining

Complete Numerical Problems & Solutions

CT 72502 (Elective I)

BEI / BEX · Year IV Part I

Institute of Engineering, Tribhuvan University

Every numerical the course sets, in one volume: **24 distinct methods** and **47 PYQ instances** drawn from **all 17 past papers (2070–2081 BS)**, plus a few extra practice problems where a likely pattern has no paper behind it yet. Each method is explained once, then every paper that used it is solved under the **question exactly as printed**, so a problem can be attempted from this PDF alone. **Every answer was independently recomputed, not copied from the source notes.**

Theory lives in the companion volume, **Data Mining — Complete Exam Notes**.

Ch	Topic	Methods	PYQ solved
2	Data Preprocessing	7	8
3	Classification	6	12
4	Association Analysis	5	14
5	Cluster Analysis	3	11
6	Anomaly / Fraud Detection	2	1
7	Advanced Applications	1	1
Total		24	47

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Ch 2 – Data Preprocessing – Numerical Problems & Solutions

7 distinct numerical methods · 8 PYQ instances solved · 2 extra practice problems. Every answer independently recomputed, not copied from the source notes.

Methods in this chapter

- **M1** Normalisation: min-max, z-score, decimal scaling 78 Bh
- **M2** PCA: principal components + proportion of variance 76 Ch
- **M3** Distance measures + distance matrix (Euclidean, Manhattan, Supremum) 80 Bh, 76 Ch
- **M4** Cosine similarity 80 Bh, 80 Ba, 76 Ch
- **M5** Binary similarity: Jaccard, SMC, symmetric dissimilarity 80 Ba, 76 Ch
- **M6** Schema design + OLAP operation sequence 78 Bh, 76 Ch
- **M7** Central tendency, quartiles, IQR and outliers NEW, no PYQ

M1. Normalisation

Method

- **Min-max** maps v into $[new_min, new_max]$, preserves relationships exactly:

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- **Z-score** uses mean $\bar{A}$ and std deviation σ_A . Use when min/max unknown, or outliers would squash a min-max result:

$$v' = \frac{v - \bar{A}}{\sigma_A} \quad \sigma_A = \sqrt{\frac{1}{n} \sum (v_i - \bar{A})^2}$$

- **Decimal scaling** moves the decimal point:

$$v' = \frac{v}{10^j}, \quad j = \text{smallest int such that } \max(|v'|) < 1$$

Two traps. (1) Z-score uses the **mean**, not the min. (2) For decimal scaling, j must make the max *strictly* less than 1, so $\max |v| = 1000$ needs $j = 4$, not $j = 3$.

Problem 1.1 2+2+2

QUESTION AS PRINTED 2078 Bhadra · Q3 · [2+2+2]

Use the following methods to normalize the data: 200, 300, 400, 600 and 1000.

- Min-max normalization by setting min = 0 and max = 1
- Z-score normalization
- Normalization by decimal scaling

Given $v = \{200, 300, 400, 600, 1000\}$, $n = 5$, $\min_A = 200$, $\max_A = 1000$.**(a) Min-max**, $\text{new_min} = 0$, $\text{new_max} = 1$

$$v' = \frac{v - 200}{1000 - 200} = \frac{v - 200}{800}$$

v	v'
200	$0/800 = 0$
300	$100/800 = 0.125$
400	$200/800 = 0.25$
600	$400/800 = 0.5$
1000	$800/800 = 1$

(c) Decimal scaling

- $\max |v| = 1000$.
- $j = 3$: $1000/10^3 = 1$, not < 1 . **Rejected.**
- $j = 4$: $1000/10^4 = 0.1 < 1$. **Use $j = 4$.**

v	$v' = v/10000$
200	0.02
300	0.03
400	0.04
600	0.06
1000	0.1

Final answer

- Min-max: 0, 0.125, 0.25, 0.5, 1
- Z-score: -1.0607, -0.7071, -0.3536, +0.3536, +1.7678
- Decimal scaling: 0.02, 0.03, 0.04, 0.06, 0.1

(b) Z-score

- Mean $\bar{A} = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$
- Deviations: -300, -200, -100, +100, +500
- $\sum (v - \bar{A})^2 = 90000 + 40000 + 10000 + 10000 + 250000 = 400000$
- $\sigma_A = \sqrt{400000/5} = \sqrt{80000} = 282.84$

v	$v' = (v - 500)/282.84$
200	$-300/282.84 = -1.0607$
300	$-200/282.84 = -0.7071$
400	$-100/282.84 = -0.3536$
600	$+100/282.84 = +0.3536$
1000	$+500/282.84 = +1.7678$

Note on σ verified/added

- Above uses the **population** std deviation, $\sqrt{\sum (v - \bar{A})^2 / n}$. This is the form Han & Kamber use, so it is the safe default.
- If the examiner wants the **sample** std deviation, divide by $n - 1$: $s = \sqrt{400000/4} = 316.23$, giving -0.9487, -0.6325, -0.3162, +0.3162, +1.5811.
- State which one you used.** Either is accepted if labelled.

M2. PCA: principal components and proportion of variance**Method**

- Start from the **covariance matrix** Σ (if given raw data: subtract the mean first, then compute Σ).
- Solve the characteristic equation $|\Sigma - \lambda I| = 0$ for the eigenvalues λ .
- Sort λ in **descending** order. λ_i = variance carried by component i .
- For each λ_i solve $(\Sigma - \lambda_i I) \mathbf{v} = 0$ for the eigenvector, then normalise it to unit length.
- Proportion of total variance explained by component i :

$$\frac{\lambda_i}{\sum_j \lambda_j}, \quad \sum_j \lambda_j = \text{trace}(\Sigma)$$

- Check:** the eigenvalues must sum to the trace. Free and catches arithmetic slips.

Problem 2.1 4

QUESTION AS PRINTED 2076 Chaitra · Q1 · [4]

Find the principal components and the proportion of the total variance explained by each when the covariance matrix of the three random variables X_1 , X_2 and X_3 is:

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Step 1: characteristic equation

- The third row/column is decoupled, so Σ splits into a 2×2 block and a 1×1 block.
- From the 1×1 block: $(2 - \lambda) = 0 \Rightarrow \boxed{\lambda = 2}$
- From the 2×2 block:

$$\begin{aligned} (1 - \lambda)(5 - \lambda) - (-2)(-2) &= 0 \\ 5 - 6\lambda + \lambda^2 - 4 &= 0 \Rightarrow \lambda^2 - 6\lambda + 1 = 0 \\ \lambda &= \frac{6 \pm \sqrt{36 - 4}}{2} = \frac{6 \pm \sqrt{32}}{2} = 3 \pm 2\sqrt{2} \end{aligned}$$

Step 2: eigenvalues, sorted

- $\lambda_1 = 3 + 2\sqrt{2} = \mathbf{5.8284}$
- $\lambda_2 = \mathbf{2}$
- $\lambda_3 = 3 - 2\sqrt{2} = \mathbf{0.1716}$
- **Check:** $5.8284 + 2 + 0.1716 = 8 = \text{trace}(\Sigma) = 1 + 5 + 2$. **Correct.**

Step 3: eigenvectors

- **For $\lambda_1 = 5.8284$:** row 1 gives $(1 - 5.8284)v_1 - 2v_2 = 0 \Rightarrow v_2 = -2.4142v_1$; row 3 gives $v_3 = 0$.
 $\mathbf{e}_1 \propto (1, -2.4142, 0)$, length $\sqrt{1 + 5.8284} = 2.6131$
 $\mathbf{e}_1 = (0.3827, -0.9239, 0)$
- **For $\lambda_2 = 2$:** rows 1 and 2 force $v_1 = v_2 = 0$, v_3 free.
 $\mathbf{e}_2 = (0, 0, 1)$
- **For $\lambda_3 = 0.1716$:** row 1 gives $0.8284v_1 = 2v_2 \Rightarrow v_2 = 0.4142v_1$; $v_3 = 0$.
 $\mathbf{e}_3 \propto (1, 0.4142, 0)$, length 1.0824
 $\mathbf{e}_3 = (0.9239, 0.3827, 0)$
- **Check:** $\mathbf{e}_1 \cdot \mathbf{e}_3 = (0.3827)(0.9239) + (-0.9239)(0.3827) = 0$
 $\rightarrow$ orthogonal, as required for a symmetric matrix.

Final answer

- PC1: $\lambda_1 = 5.8284$, $\mathbf{e}_1 = (0.3827, -0.9239, 0)$, **72.86%**
- PC2: $\lambda_2 = 2$, $\mathbf{e}_2 = (0, 0, 1)$, **25.00%**
- PC3: $\lambda_3 = 0.1716$, $\mathbf{e}_3 = (0.9239, 0.3827, 0)$, **2.14%**

Step 4: proportion of variance

PC	λ_i	$\lambda_i/8$
PC1	5.8284	72.86%
PC2	2.0000	25.00%
PC3	0.1716	2.14%
Total	8.0000	100%

Interpretation

- PC1 alone carries $\approx 73\%$ of the variance.
- PC1 + PC2 carry **97.86%** $\rightarrow$ dropping PC3 loses only 2.14%.
- So 3 dims reduce to **2** with almost no information loss.

M3. Distance measures and the distance matrix**Method**

- All three are Minkowski with a different r :

$$\text{dist}(p, q) = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{1/r}$$

- $r = 1$ **Manhattan:** $\sum |p_k - q_k|$
- $r = 2$ **Euclidean:** $\sqrt{\sum (p_k - q_k)^2}$
- $r \rightarrow \infty$ **Supremum** (L_∞ , Chebyshev): $\max_k |p_k - q_k|$
- A **distance matrix** is symmetric with a zero diagonal, so only the lower triangle needs computing.

Warm-up (Han & Kamber Example 2.19–2.20)

PROBLEM not a PYQ · Han & Kamber Example 2.19–2.20 verified/added

Given the two objects $x_1 = (1, 2)$ and $x_2 = (3, 5)$, compute the Euclidean, Manhattan and Supremum distances between them.

Smallest case that shows all three at once. Worth memorising as a check.

- $x_1 = (1, 2), x_2 = (3, 5)$. Differences: 2 and 3.
- Euclidean = $\sqrt{2^2 + 3^2} = \sqrt{13} = \mathbf{3.61}$
- Manhattan = $2 + 3 = \mathbf{5}$
- Supremum = $\max(2, 3) = \mathbf{3}$
- **Ordering:** supremum $\leq$ Euclidean $\leq$ Manhattan. Always true. Use it to catch slips.

Problem 3.1 2

QUESTION AS PRINTED 2076 Chaitra · Q2 (a) · [2]

Given the following points compute the distance matrix using the Manhattan and the Supremum distance.

Points	X	Y
P1	6	3
P2	2	2
P3	3	4

Manhattan (L_1)

- $d(P_1, P_2) = |6 - 2| + |3 - 2| = 4 + 1 = 5$
- $d(P_1, P_3) = |6 - 3| + |3 - 4| = 3 + 1 = 4$
- $d(P_2, P_3) = |2 - 3| + |2 - 4| = 1 + 2 = 3$

Supremum (L_∞)

- $d(P_1, P_2) = \max(4, 1) = 4$
- $d(P_1, P_3) = \max(3, 1) = 3$
- $d(P_2, P_3) = \max(1, 2) = 2$

	P_1	P_2	P_3
P_1	0	5	4
P_2	5	0	3
P_3	4	3	0

	P_1	P_2	P_3
P_1	0	4	3
P_2	4	0	2
P_3	3	2	0

- **Sanity check:** supremum $\leq$ Euclidean $\leq$ Manhattan, always. Holds here ($4 \leq 5, 3 \leq 4, 2 \leq 3$).

Problem 3.2 3

QUESTION AS PRINTED 2080 Bhadra · Q1 (b) · [3+3+3]

How do similarity / dissimilarity is calculated? Find the cosine similarity between Object-2 and 4. Also calculate the Euclidean distance between object 1, 3 and object 1, 4.

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

The cosine-similarity part of this same question is Problem 4.1 below.

Step-by-step

- $d(O_1, O_3)$: differences $(4 - 7, 56 - 58, 7 - 6, 10 - 9) = (-3, -2, 1, 1)$

$$= \sqrt{9 + 4 + 1 + 1} = \sqrt{15} = \mathbf{3.873}$$
- $d(O_1, O_4)$: differences $(4 - 9, 56 - 55, 7 - 7, 10 - 12) = (-5, 1, 0, -2)$

$$= \sqrt{25 + 1 + 0 + 4} = \sqrt{30} = \mathbf{5.477}$$

Final answer

$d(O_1, O_3) = 3.873$, $d(O_1, O_4) = 5.477$. Object 1 is closer to object 3 than to object 4.

M4. Cosine similarity

Method

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}, \quad d_1 \cdot d_2 = \sum_k d_{1k} d_{2k}, \quad \|d\| = \sqrt{\sum_k d_k^2}$$

- Measures **angle only**, ignores magnitude → used for documents.
- Always lay out 3 lines: dot product, then each length, then divide.

Typo in your class notes verified/added : DBK's Lecture-2 cosine example prints $\|d_2\| = (6)^{0.5} = 2.245$. In fact $\sqrt{6} = \mathbf{2.449}$. His final answer 0.3150 is correct and only follows from $2.449 (5/(6.481 \times 2.449) = 0.315)$, so 2.245 is a slip. Copy the method, not that number.

Problem 4.1 3

QUESTION AS PRINTED 2080 Bhadra · Q1 (b) · [3+3+3]

How do similarity / dissimilarity is calculated? Find the cosine similarity between Object-2 and 4. Also calculate the Euclidean distance between object 1, 3 and object 1, 4.

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

The Euclidean-distance part of this same question is Problem 3.2 above.

Given

$$O_2 = (3, 53, 8, 11), \quad O_4 = (9, 55, 7, 12).$$

Step-by-step

- Dot product: $3(9) + 53(55) + 8(7) + 11(12) = 27 + 2915 + 56 + 132 = 3130$
- $\|O_2\| = \sqrt{9 + 2809 + 64 + 121} = \sqrt{3003} = 54.800$
- $\|O_4\| = \sqrt{81 + 3025 + 49 + 144} = \sqrt{3299} = 57.437$
- $\cos = \frac{3130}{54.800 \times 57.437} = \frac{3130}{3147.5}$

Final answer

$$\cos(O_2, O_4) = \mathbf{0.9944}.$$

- Nearly 1 → the two objects point almost the same direction.
- **Why so high:** *Weight* (≈ 50) dwarfs the other attributes and dominates the dot product. **Normalise before using cosine** on mixed-scale attributes, or the answer means very little. verified/added

Problem 4.2 1

QUESTION AS PRINTED 2080 Baishakh · Q2 (b)(iv) · [1]

Find the Cosine similarity between documents $d_1 = (4, 1, 2, 0, 2, 0, 0)$ and $d_2 = (2, 1, 3, 0, 1, 1, 1)$. (the rest of Q2 b is Problem 5.2 below)

- Dot: $8 + 1 + 6 + 0 + 2 + 0 + 0 = 17$
- $\|d_1\| = \sqrt{16 + 1 + 4 + 0 + 4 + 0 + 0} = \sqrt{25} = 5$
- $\|d_2\| = \sqrt{4 + 1 + 9 + 0 + 1 + 1 + 1} = \sqrt{17} = 4.1231$
- $\cos = 17/(5 \times 4.1231) = 17/20.6155$

Final answer

$$\mathbf{0.8246}$$

Problem 4.3 1

QUESTION AS PRINTED 2076 Chaitra · Q2 (b) · [1]

Given the following two vectors compute the Cosine similarity between them.

$$D_1 = [4 \ 0 \ 2 \ 0 \ 1] \quad D_2 = [2 \ 0 \ 0 \ 2 \ 2]$$

- Dot: $8 + 0 + 0 + 0 + 2 = 10$
- $\|D_1\| = \sqrt{16 + 0 + 4 + 0 + 1} = \sqrt{21} = 4.5826$
- $\|D_2\| = \sqrt{4 + 0 + 0 + 4 + 4} = \sqrt{12} = 3.4641$
- $\cos = 10 / (4.5826 \times 3.4641) = 10 / 15.8745$

Final answer

0.6300

M5. Binary similarity: Jaccard, SMC, symmetric dissimilarity

Method

- Encode each object as a 0/1 vector, then count over all attribute positions:
 - M_{11} : both 1 M_{00} : both 0
 - M_{10} : $p = 1, q = 0$ M_{01} : $p = 0, q = 1$
- **SMC** (symmetric binary, 0-0 counts): $\frac{M_{11} + M_{00}}{M_{11} + M_{00} + M_{10} + M_{01}}$
- **Jaccard** (asymmetric binary, 0-0 ignored): $\frac{M_{11}}{M_{11} + M_{10} + M_{01}}$
- **Symmetric binary dissimilarity**: $d = \frac{M_{10} + M_{01}}{M_{11} + M_{00} + M_{10} + M_{01}} = 1 - SMC$
- **Nominal (non-binary) dissimilarity**: $d = (p - m) / p$, $m = \text{matches}$, $p = \text{total attrs}$.

Problem 5.1 2

QUESTION AS PRINTED 2076 Chaitra · Q2 (c) · [2]

Given the following two binary vectors compute the Jaccard similarity and Simple Matching Coefficient.

$$P = [0 \ 0 \ 1 \ 1 \ 0 \ 1] \quad Q = [1 \ 1 \ 1 \ 1 \ 0 \ 1]$$

Step-by-step: tabulate position by position

Position	1	2	3	4	5	6
P	0	0	1	1	0	1
Q	1	1	1	1	0	1
Category	M_{01}	M_{01}	M_{11}	M_{11}	M_{00}	M_{11}

- $M_{11} = 3, \quad M_{01} = 2, \quad M_{10} = 0, \quad M_{00} = 1 \quad (\text{total} = 6 \checkmark)$
- $J = \frac{3}{3 + 0 + 2} = \frac{3}{5}$
- $SMC = \frac{3 + 1}{6} = \frac{4}{6}$

Final answer

$J = 0.6, \quad SMC = 0.667$. SMC is higher because it credits the one 0-0 match that Jaccard discards.

Problem 5.2 2+2+2

QUESTION AS PRINTED 2080 Baishakh · Q2 (b) · [2+2+2+1]

Consider the following table:

Name	Gender	Eyecolor	Haircolor	Test-1	Test-2	Fever	Cough
Ram	M	Black	Gray	P	N	P	N
Laxmi	F	Blue	Black	P	P	N	N
Shyam	M	Blue	Gray	N	P	N	P

- i) Calculated Jaccard Coefficient of $Sim_{jaccard}$ (Ram, Laxmi) for asymmetric binary attributes.
- ii) Dissimilarity of symmetric binary attributes d (Laxmi, Shyam).

- iii) Find the Simple matching coefficients of SMC (Ram, Shyam).
- iv) Find the Cosine similarity between documents $d_1 = (4, 1, 2, 0, 2, 0, 0)$ and $d_2 = (2, 1, 3, 0, 1, 1, 1)$ – solved as Problem 4.2 above.

Step 0: split the attributes (the whole question turns on this)

- **Asymmetric binary** = Test-1, Test-2, Fever, Cough. Encode $P = 1, N = 0$. A shared “N” is uninformative, which is exactly what “asymmetric” means.
- **Symmetric** = Gender, Eyecolor, Haircolor. Both/all states are equally meaningful.
- Encoded: Ram = (1, 0, 1, 0), Laxmi = (1, 1, 0, 0), Shyam = (0, 1, 0, 1).

(i) Jaccard(Ram, Laxmi)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Laxmi	1	1	0	0
Cat.	M_{11}	M_{01}	M_{10}	M_{00}

- $M_{11} = 1, M_{01} = 1, M_{10} = 1, M_{00} = 1$
- $J = \frac{1}{1 + 1 + 1} = \frac{1}{3}$

Answer

$J = 0.333$

(iii) SMC(Ram, Shyam)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Shyam	0	1	0	1
Cat.	M_{10}	M_{01}	M_{10}	M_{01}

- $M_{11} = 0, M_{00} = 0, M_{10} = 2, M_{01} = 2$
- $SMC = \frac{0+0}{4} = 0$

Answer

$SMC = 0$ (they disagree on every test).

(ii) $d(\text{Laxmi, Shyam})$, symmetric binary attributes

- Across these 3 people each of Gender / Eyecolor / Haircolor takes exactly **2 states**, and neither state is an “absence” → all 3 are **symmetric binary**.
- Encode: Gender $M=1$ · Eyecolor Black=1 · Haircolor Gray=1.

	Gender	Eye	Hair
Laxmi (F, Blue, Black)	0	0	0
Shyam (M, Blue, Gray)	1	0	1
Category	s	t	s

- $q = 0, r = 0, s = 2, t = 1$ (total $p = 3$ ✓)
- Symmetric binary dissimilarity keeps t :

$$d = \frac{r + s}{q + r + s + t} = \frac{0 + 2}{3} = \frac{2}{3}$$

Answer

$d = 0.667$

Cross-check: for binary attrs the symmetric formula reduces to the nominal mismatch ratio $(p - m)/p = (3 - 1)/3 = 2/3$. Same answer by both routes, so the encoding choice is safe.

verified/added

M6. Schema design + OLAP operation sequence

Method

1. Put the **measures** in a central **fact table**, plus one foreign key per dimension.
2. Draw one **dimension table** per dimension around it.
 - **Star** → leave dimension tables flat (de-normalised).
 - **Snowflake** → normalise the hierarchies into sub-tables (location → city → province).
3. For the OLAP sequence, work from the base cuboid and ask, per dimension:
 - Need a coarser level? → **roll-up**
 - Need a finer level? → **drill-down**
 - Restricting 1 dimension? → **slice**. Restricting ≥ 2 ? → **dice**
 - Dimension not wanted at all? → **roll-up to all** (dimension reduction)
4. n dimensions → 2^n cuboids in the lattice.

Problem 6.1 3+3

QUESTION AS PRINTED 2078 Bhadra · Q2 · [3+3]

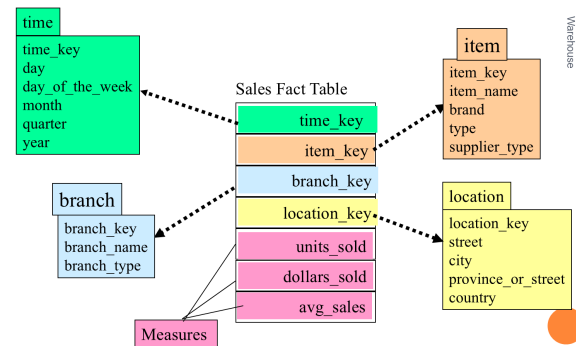
Suppose that a data warehouse consists of the four dimensions data, spectator, location, and game, and the two measures count and charge, where charge is the fare that a spectator pays when watching a game on a given date. Spectators may be students, adults or seniors, with each category having its own charge rate.

- Draw a star schema diagram for the data warehouse.
- Starting with the base cuboid [data, spectator, location, game], what specific OLAP operations should you perform in order to list the total charge paid by student spectators at Dashrath Stadium in 2021?

(a) Star schema

- **Fact table** Sales: date_key, spectator_key, location_key, game_key, count, charge.
- **Dimension tables** (flat, one hop from the fact table):
 - date(date_key, day, month, quarter, year)
 - spectator(spectator_key, name, status, phone, address) where status ∈ {student, adult, senior}
 - location(location_key, name, street, city, province, country)
 - game(game_key, name, description, sport_type)
- 4 dimensions → $2^4 = 16$ cuboids in the lattice.

EXAMPLE OF STAR SCHEMA



(b) OLAP operation sequence

1. **Roll-up** on *date* from date_key to year.
 2. **Roll-up** on *spectator* from spectator_key to status.
 3. **Roll-up** on *location* from location_key to location_name.
 4. **Roll-up** on *game* to all (the question does not split by game).
 5. **Dice** with year = 2021 **and** status = student **and** location_name = ``Dashrath Stadium``.
 6. Read off the charge measure (summed).
- **Dice, not slice** → 3 dimensions are being restricted at once.

Problem 6.2 3+3

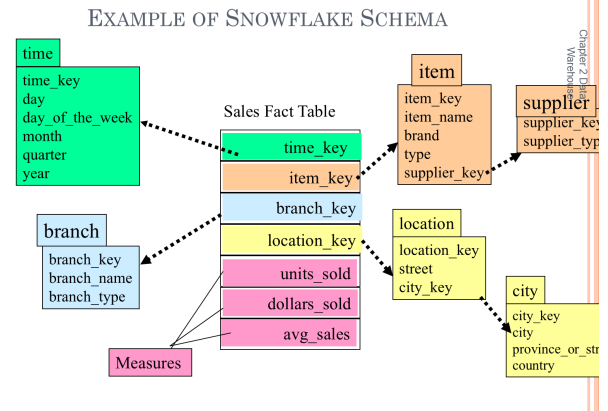
QUESTION AS PRINTED 2076 Chaitra · Q3 · [3+3]

Suppose that a data warehouse for a sales company consists of five dimensions: *time*, *location*, *supplier*, *brand*, and *product*, and two measures: *count* and *price*.

- Draw a *snowflake schema* diagram for the data warehouse.
- Starting with the base cuboid [*time*, *location*, *supplier*, *brand*, *product*], what specific OLAP operations should one perform in order to list the total *count* for a certain *brand* for each *state* per year (assume *location* has three levels: *country*, *state*, *city*; and assume *time* has three levels: *year*, *month*, *day*)?

(a) Snowflake schema

- **Fact table** Sales: time_key, location_key, supplier_key, brand_key, product_key, count, price.
- **Normalised hierarchies** (this is what makes it a snowflake, not a star):
 - time(time_key, day, month_key) → month(month_key, month, year)
 - location(location_key, city_key) → city(city_key, city, state_key) → state(state_key, state, country)
 - product(product_key, name, brand_key) → brand(branch_key, brand_name)
 - supplier(supplier_key, supplier_name, supplier_type)
- 5 dimensions → $2^5 = 32$ cuboids.

**(b) OLAP operation sequence**

1. Roll-up on *time* from day to year.
2. Roll-up on *location* from city to state.
3. Roll-up on *supplier* to all (not wanted).
4. Roll-up on *product* to brand level.
5. Slice on brand = the given brand (1 dimension restricted).
6. Result: cuboid [year, state] holding total count.

M7. Central tendency, quartiles, IQR and outliers

NEW Not in any of the 17 papers. Added because Notes by PC/unit 2 Numerical.docx and Untitled spreadsheet.xlsx (2025-26) are built entirely around it. Theory in Ch 2, §2.5.

Method

- **Grouped median** = $L_1 + \left(\frac{N/2 - (\sum freq)_l}{freq_{median}} \right) \times width$. Find the median interval first: the one where cumulative frequency first reaches $N/2$.
- **Mode from mean and median** $\approx 3 median - 2 mean$ (unimodal, moderately skewed).
- **Quartiles**, interpolated convention: position of $Q_k = \frac{k(n+1)}{4}$, then interpolate between the neighbouring sorted values.
- **Outlier fences**: below $Q_1 - 1.5 IQR$ or above $Q_3 + 1.5 IQR$.

Problem 7.1 – grouped median, 100 employees 4

PROBLEM not a PYQ • PC's notes • [4]

Suppose the salaries of 100 employees of a company are grouped into \$10,000 intervals as shown in the table below. Find the median salary.

Salary (\$)	Freq	Cum freq
10,000–20,000	8	8
20,000–30,000	15	23
30,000–40,000	20	43
40,000–50,000	32	75
50,000–60,000	18	93
60,000–70,000	7	100

Solution

1. $N = 100 \rightarrow N/2 = 50$. Need the 50th observation.
2. Cumulative frequency passes 50 inside 40k–50k (43 before it, 75 after) → that is the median interval.
3. $L_1 = 40,000$, $(\sum freq)_l = 43$,
 $freq_{median} = 75 - 43 = 32$, $width = 10,000$.
4. $median = 40,000 + \frac{50-43}{32} \times 10,000 = 40,000 + \frac{7}{32} \times 10,000$

Answer

median = \$42,187.50

Problem 7.2 – same method, smaller table 3**PROBLEM** not a PYQ · PC's notes · [3]

Salaries (in \$1000s) of 60 employees fall in the intervals 10–20, 20–30, 30–40, 40–50, 50–60 and 60–70 with frequencies 5, 8, 12, 20, 10 and 5 respectively. Find the median salary.

- $N = 5 + 8 + 12 + 20 + 10 + 5 = 60 \rightarrow N/2 = 30$.
- Cumulative before 40–50 = $5 + 8 + 12 = 25$; $freq_{median} = 20$, $L_1 = 40$, $width = 10$.
- $median = 40 + \frac{30-25}{20} \times 10 = 40 + 2.5 = \mathbf{42.5}$ (i.e. \$42,500).

Problem 7.3 – mode from mean and median 2**PROBLEM** not a PYQ · PC's notes · [2]

For a unimodal, moderately skewed distribution the mean is 60 and the median is 55. Estimate the mode, and state what the result says about the skew.

- $mean - mode \approx 3(mean - median) \Rightarrow mode \approx 3median - 2mean$.
- $mode \approx 3(55) - 2(60) = 165 - 120 = \mathbf{45}$.
- **Sanity check:** $mode < median < mean \rightarrow$ positively skewed. Consistent.

Problem 7.4 – full descriptive summary of one dataset 2+2+2+2**PROBLEM** not a PYQ · Han & Kamber Example 2.6 · [2+2+2+2]

Suppose that the data for analysis is the salaries (in \$1000s) of 12 employees, already sorted in increasing order:

30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

Find (a) the mean, median and mode; (b) the quartiles Q_1 and Q_3 and the IQR; (c) the five-number summary; (d) any outliers using the $1.5 \times IQR$ rule.

(a) Central tendency

- $\sum x = 696 \rightarrow mean = \frac{696}{12} = \mathbf{58}$.
- n even $\rightarrow median = \frac{52+56}{2} = \mathbf{54}$.
- 52 and 70 each occur twice $\rightarrow$ **bimodal**, modes **52** and **70**.
- $midrange = \frac{30+110}{2} = \mathbf{70}$.
- $mode < median < mean$ is broken here by the bimodality; the single outlier 110 is what pulls the mean above the median.

(b) Quartiles, positional convention (Han & Kamber)

- Take the 3rd, 6th, 9th sorted values $\rightarrow Q_1 = 47$, $Q_2 = 52$, $Q_3 = 63$.
- $IQR = 63 - 47 = \mathbf{16}$.

(c) Quartiles, interpolated convention (PC's class notes)

- Q_1 at $\frac{1(13)}{4} = 3.25 \rightarrow 47 + 0.25(50 - 47) = \mathbf{47.75}$
- Q_2 at $\frac{2(13)}{4} = 6.5 \rightarrow 52 + 0.5(56 - 52) = \mathbf{54}$
- Q_3 at $\frac{3(13)}{4} = 9.75 \rightarrow 63 + 0.75(70 - 63) = \mathbf{68.25}$
- $IQR = 68.25 - 47.75 = \mathbf{20.5}$

(d) Five-number summary and outliers

- Five-number summary (interpolated): **30, 47.75, 54, 68.25, 110**.
- Fences: $47.75 - 1.5(20.5) = 17.0$ and $68.25 + 1.5(20.5) = 99.0$.
- $110 > 99 \rightarrow \mathbf{110}$ is a suspected outlier. Nothing below 17.
- **Cross-check with the other convention:** fences $47 - 24 = 23$ and $63 + 24 = 87$. **110 is still the only outlier** $\rightarrow$ the verdict does not depend on the convention, so either is safe in the exam as long as you name it.

(e) Boxplot

- Box from Q_1 to Q_3 , line at 54, whiskers to 30 and to the largest non-outlier (70), and **110 plotted as a lone point**.

Extra practice (patterns not yet fully covered)**Practice A – binning for noise smoothing**

Binning is examined in prose 72 Ka, 76 Ash but never as a numerical. **This is DBK's own worked example** (Lecture-1, "Binning Methods for Data Smoothing"), so it is the likeliest form any future numerical would take. Answers below independently recomputed and they match his slide exactly.

PROBLEM not a PYQ · DBK Lecture-1 worked example verified/added

The following price data has already been sorted:

4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

Partition the data into **equal-depth** bins of depth 4, then smooth the noise by (i) bin means and (ii) bin boundaries.

Solution

- 12 values, depth 4 $\rightarrow$ 3 bins.
 - Bin 1: 4, 8, 9, 15 Bin 2: 21, 21, 24, 25 Bin 3: 26, 28, 29, 34
- **(i) Smoothing by bin means**
 - Bin 1 mean = $36/4 = 9 \rightarrow 9, 9, 9, 9$
 - Bin 2 mean = $91/4 = 22.75 \approx 23 \rightarrow 23, 23, 23, 23$
 - Bin 3 mean = $117/4 = 29.25 \approx 29 \rightarrow 29, 29, 29, 29$
- **(ii) Smoothing by bin boundaries** (each value $\rightarrow$ nearer of min/max)
 - Bin 1 (min 4, max 15): 4, 8, 9, 15 $\rightarrow 4, 4, 4, 15$ (8 and 9 are nearer 4)
 - Bin 2 (min 21, max 25): 21, 21, 24, 25 $\rightarrow 21, 21, 25, 25$
 - Bin 3 (min 26, max 34): 26, 28, 29, 34 $\rightarrow 26, 26, 26, 34$
- **Equal-width** for contrast: range = $34 - 4 = 30$, $N = 3 \rightarrow W = (B - A)/N = 10$. Intervals $[4, 14)$, $[14, 24)$, $[24, 34]$ $\rightarrow$ counts 3, 3, 6. **Unequal counts: that is the difference from equal-depth.**

Practice B — min-max into a range other than [0,1]

Every PYQ so far uses $[0, 1]$. The figures below are Han & Kamber 3rd ed Example 3.4, so they are the canonical ones. verified/added

PROBLEM not a PYQ · Han & Kamber 3rd ed Example 3.4 verified/added

Suppose that the minimum and maximum values for the attribute *income* are \$12,000 and \$98,000 respectively. Normalise **\$73,600** by min-max normalization, first to the range $[0, 1]$ and then to the range $[-1, 1]$.

Solution

- To $[0, 1]$: $v' = \frac{73600 - 12000}{98000 - 12000}(1 - 0) + 0 = \frac{61600}{86000} = \mathbf{0.716}$
- To $[-1, 1]$: $v' = \frac{61600}{86000}(1 - (-1)) + (-1) = 0.716(2) - 1 = \mathbf{0.432}$
- **Check:** the value sits 71.6% of the way up the range, so in $[-1, 1]$ it must land slightly above the midpoint 0. It does.

Ch 3 – Classification – Numerical Problems & Solutions

6 distinct methods · 12 PYQ instances solved · every answer independently recomputed. Decision-tree and Naive-Bayes figures cross-checked against Han & Kamber 3rd ed Examples 8.1–8.4, which use the same dataset the IOE papers copy.

Methods

- **M1** Decision tree by information gain (ID3) 79 Bh, 78 Bh, 76 Ch
- **M2** Gini index 75 Ch, 71 Ch
- **M3** Naive Bayes classification 72 Ch, 79 Bh
- **M4** k -Nearest Neighbour 80 Bh
- **M5** Confusion matrix metrics **8 papers**
- **M6** ANN gradients (backpropagation) 76 Ch

M1. Decision tree by information gain (ID3)

Method

1. Compute $Info(D) = -\sum p_i \log_2 p_i$ for the whole set.
 2. For each attribute A : split D , compute $Info_A(D) = \sum \frac{|D_j|}{|D|} Info(D_j)$, then $Gain(A) = Info(D) - Info_A(D)$.
 3. **Highest gain wins** → becomes the node's test.
 4. Recurse on each branch. Stop when a branch is **pure** ($Info = 0$).
- **Time-savers**: a pure partition contributes 0; a 50/50 partition contributes exactly 1; two partitions with the same class ratio have the same $Info$, so compute once.
 - $\log_2 x = \ln x / \ln 2$. Keep 4 decimals until the end.

Problem 1.1 10

QUESTION AS PRINTED 2079 Bhadra · Q3 · [10]

Draw decision tree for the given data using ID3 algorithm.

Age	Income	Student	Credit_Rating	Buy's_Computer
Youth	High	No	Fair	No
Youth	High	No	Excellent	No
Middle_Aged	High	No	Fair	Yes
Senior	Medium	No	Fair	Yes
Senior	Low	Yes	Fair	Yes
Senior	Low	Yes	Excellent	No
Middle_Aged	Low	Yes	Excellent	Yes
Youth	Medium	No	Fair	No
Youth	Low	Yes	Fair	Yes
Senior	Medium	Yes	Fair	Yes
Youth	Medium	Yes	Excellent	Yes
Middle_Aged	Medium	No	Excellent	Yes
Middle_Aged	High	Yes	Fair	Yes
Senior	Medium	No	Excellent	No

Given

14 tuples. Class counts: **Yes = 9, No = 5**.

Step 1: entropy of the whole set

$$Info(D) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.4098 + 0.5305 = \mathbf{0.940}$$

Step 2: gain of each attribute

Attribute	Partition (Yes, No)	$Info_A(D)$	Gain
Age	Youth (2,3) · Middle_Aged (4,0) · Senior (3,2)	$\frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.694$	0.246
Income	High (2,2) · Medium (4,2) · Low (3,1)	$\frac{4}{14}(1.0) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811) = 0.911$	0.029
Student	Yes (6,1) · No (3,4)	$\frac{7}{14}(0.592) + \frac{7}{14}(0.985) = 0.788$	0.151
Credit_Rating	Fair (6,2) · Excellent (3,3)	$\frac{8}{14}(0.811) + \frac{6}{14}(1.0) = 0.892$	0.048

– **Age has the highest gain → Age is the root.**

– Note Middle_Aged is (4, 0): **pure**, so that branch is immediately a leaf “Yes”. That purity is what makes Age win.

Step 3: recurse on the Youth branch

(5 tuples: 2 Yes, 3 No, $Info = 0.971$)

- Student = Yes → (2 Yes, 0 No) Student = No → (0 Yes, 3 No)
- Both pure → $Info_{Student} = 0$ → $Gain = 0.971$, the maximum possible.
- **Student** is the test under Youth.

Step 4: recurse on the Senior branch

(5 tuples: 3 Yes, 2 No, $Info = 0.971$)

- Credit = Fair → (3 Yes, 0 No) Credit = Excellent → (0 Yes, 2 No)
- Both pure → $Gain = 0.971$. **Credit_Rating** is the test under Senior.

Final tree

- **Age?**
 - Youth → **Student?**
 - no → **No**
 - yes → **Yes**
 - Middle_Aged → **Yes**
 - Senior → **Credit_Rating?**
 - fair → **Yes**
 - excellent → **No**
- Tree has **5 leaves**, depth 2, and classifies all 14 training tuples correctly.

8.2 Decision Tree Induction 331

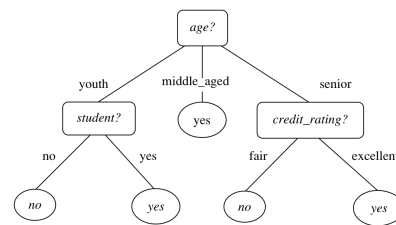


Figure 8.2 A decision tree for the concept *buys.computer*, indicating whether an *AlIElectronics* customer is likely to purchase a computer. Each internal (nonleaf) node represents a test on

The same tree as printed in Han & Kamber 3rd ed, Fig 8.2. Our independently computed gains match their Example 8.1 exactly.

Problem 1.2 8

QUESTION AS PRINTED 2078 Bhadra · Q4 · [8]

Construct a decision tree for the following data set using information gain.

Predict the class label for a data point with values <Female, 2, standard, high>

Gender	Car ownership	Travel cost	Income level	Transport mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Female	1	Cheap	Medium	Train
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

Given

Classes: Bus = 4, Car = 3, Train = 3. $n = 10$.

$$Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{3}{10} \log_2 \frac{3}{10} = 0.5288 + 0.5211 + 0.5211 = \mathbf{1.571}$$

Travel cost

Cheap (4 Bus, 0 Car, 1 Train) · Expensive (0,3,0) · Standard (0,0,2)

- $Info(\text{Cheap}) = -\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5} = 0.722$
- Expensive and Standard are both **pure** $\rightarrow Info = 0$
- $Info_{TravelCost}(D) = \frac{5}{10}(0.722) + \frac{3}{10}(0) + \frac{2}{10}(0) = 0.361$
- $Gain = 1.571 - 0.361 = \mathbf{1.210}$
- Every other attribute scores far lower (Gender 0.125, Car ownership 0.534, Income level 0.971), so **Travel cost is the root** – two of its three branches are pure.
- Only the *Cheap* branch needs splitting: 5 tuples (4 Bus, 1 Train). Splitting it on **Gender** gives Male (3 Bus, 0 Train) and Female (1 Bus, 1 Train); splitting on **Car ownership** gives 0 (2 Bus), 1 (2 Bus, 1 Train). Neither is perfect; Gender has the higher gain, then Car ownership finishes the Female sub-branch.

Prediction for <Female, 2, standard, high>

- Travel cost = **standard** $\rightarrow$ that branch is **pure Train**.
- **No further tests are needed.** **Answer: Train.**

M2. Gini index**Method**

$$Gini(D) = 1 - \sum_i p_i^2 \quad Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2) \quad \Delta Gini = Gini(D) - Gini_A(D)$$

- **Gini uses BINARY splits only.** For an attribute with v values you must try all $2^{v-1} - 1$ ways of splitting the values into 2 subsets and keep the best.
- Pick the largest $\Delta Gini$ (equivalently the smallest $Gini_A$).

Worked example (same 14-tuple dataset) 6

QUESTION AS PRINTED 2075 Chaitra · Q3 · [6+4]

Define Decision Tree Classifier with Gini-Index with suitable example. How can you handle overfitting in Decision Tree?

QUESTION AS PRINTED 2071 Chaitra · Q3 · [10]

What is a decision tree? Explain Gini Index with suitable example.

Neither paper prints a dataset – the “suitable example” is yours to supply. The 14-tuple *buys_computer* set below is the safe one to reuse. verified/added

- $Gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 1 - 0.4133 - 0.1276 = \mathbf{0.459}$
- Split **Income** as {low, medium} vs {high}:
 - $D_1 = 10$ tuples (7 Yes, 3 No): $Gini = 1 - (0.7)^2 - (0.3)^2 = 0.420$
 - $D_2 = 4$ tuples (2 Yes, 2 No): $Gini = 1 - (0.5)^2 - (0.5)^2 = 0.500$
 - $Gini_{Income}(D) = \frac{10}{14}(0.420) + \frac{4}{14}(0.500) = 0.300 + 0.143 = 0.443$
 - $\Delta Gini = 0.459 - 0.443 = \mathbf{0.016}$
- Split **Age** as {youth, senior} vs {middle_aged}:
 - $D_1 = 10$ tuples (5 Yes, 5 No): $Gini = 0.500$
 - $D_2 = 4$ tuples (4 Yes, 0 No): $Gini = \mathbf{0}$ (pure)
 - $Gini_{Age}(D) = \frac{10}{14}(0.500) + \frac{4}{14}(0) = 0.357$
 - $\Delta Gini = 0.459 - 0.357 = \mathbf{0.102} \rightarrow \text{best, so Age is chosen}$
- Same winner as information gain, which is the usual (not guaranteed) outcome.

M3. Naive Bayes classification

Method

1. Priors: $P(C_i) = |C_{i,D}|/|D|$.
2. For each attribute value in X : $P(x_k|C_i)$ from counts within class C_i .
3. $P(X|C_i) = \prod_k P(x_k|C_i)$.
4. Compare $P(X|C_i)P(C_i)$ across classes; **largest wins**.
 - **Never divide by $P(X)$** – it is identical for every class and cancels.
 - Any zero count $\rightarrow$ apply **Laplacian correction** $\frac{n_k+1}{|C_{i,D}|+k}$.
 - Lay the work out as a small table of conditional probabilities; it earns method marks even if arithmetic slips.

Problem 3.1 10

QUESTION AS PRINTED 2072 Chaitra · Q3 · [10]

Predict Class label using naive Bayesian classifier for $X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit-rating} = \text{fair})$ using the following data set.

RID	Age	Income	Student	Credit-rating	Class Buy computer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle-age	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle-age	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle-age	Medium	No	Excellent	Yes
13	Middle-age	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

Given

14 tuples. $P(\text{Yes}) = 9/14 = 0.643$, $P(\text{No}) = 5/14 = 0.357$.

Conditional probabilities

	Yes	No
$P(\text{age} = \text{youth})$	$2/9 = 0.222$	$3/5 = 0.600$
$P(\text{income} = \text{med})$	$4/9 = 0.444$	$2/5 = 0.400$
$P(\text{student} = \text{yes})$	$6/9 = 0.667$	$1/5 = 0.200$
$P(\text{credit} = \text{fair})$	$6/9 = 0.667$	$2/5 = 0.400$

Combine

- $P(X|\text{Yes}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = \mathbf{0.0439}$
- $P(X|\text{No}) = 0.600 \times 0.400 \times 0.200 \times 0.400 = \mathbf{0.0192}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.0439 \times 0.643 = \mathbf{0.028}$
- $P(X|\text{No})P(\text{No}) = 0.0192 \times 0.357 = \mathbf{0.007}$

Final answer

$0.028 > 0.007 \rightarrow \text{buys_computer} = \text{Yes}$.

- Matches Han & Kamber 3rd ed Example 8.4.

Problem 3.2 6

QUESTION AS PRINTED 2079 Bhadra · Q4 · [6]

Suppose you have a test record “ X = (Home Owner = No, Marital Status = Married, Income = \$120K)”. Your job is to classify this record using Naive Bayesian Classification. Use the following table for your calculations.

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given

Defaulted: **No = 7, Yes = 3**. $P(\text{No}) = 0.7$, $P(\text{Yes}) = 0.3$.

Step 1: categorical attributes

- $P(\text{HO} = \text{No} | \text{No}) = 4/7 = 0.571$ $P(\text{HO} = \text{No} | \text{Yes}) = 3/3 = 1.0$
- $P(\text{Married} | \text{No}) = 4/7 = 0.571$ $P(\text{Married} | \text{Yes}) = 0/3 = 0 \leftarrow \text{zero!}$
- Apply **Laplacian correction** (Marital Status has $k = 3$ values): $P(\text{Married} | \text{Yes}) = \frac{0+1}{3+3} = 0.167$; and $P(\text{HO} = \text{No} | \text{Yes}) = \frac{3+1}{3+2} = 0.8$.

Step 2: continuous attribute (Income), Gaussian

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Class **No**: incomes 125, 100, 70, 120, 60, 220, 75 $\rightarrow \mu = 110, \sigma = 54.54$

$$g(120) = \frac{1}{\sqrt{2\pi}(54.54)} e^{-\frac{(10)^2}{2(2975)}} = 0.00719$$

- Class **Yes**: incomes 95, 85, 90 $\rightarrow \mu = 90, \sigma = 5$

$$g(120) = \frac{1}{\sqrt{2\pi}(5)} e^{-\frac{(30)^2}{2(25)}} = 0.0798 e^{-18} \approx 1.2 \times 10^{-9}$$

- **The Yes class has a tiny spread ($\sigma = 5$) around \$90K, so \$120K is 6 standard deviations out** – effectively impossible under that model.

Step 3: combine

- $P(X|\text{No})P(\text{No}) = 0.571 \times 0.571 \times 0.00719 \times 0.7 = \mathbf{1.64 \times 10^{-3}}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.8 \times 0.167 \times 1.2 \times 10^{-9} \times 0.3 \approx \mathbf{4.8 \times 10^{-11}}$

Final answer

Defaulted Borrower = No (larger by roughly 10^7).

M4. k -Nearest Neighbour

Method

1. Compute the Euclidean distance from the query to **every** training tuple.
 2. Sort ascending, take the k smallest.
 3. **Majority class** among those k is the prediction.
- **Show the full distance table** – that is where the marks are, not the final label.
 - You may compare squared distances to save time, but write the roots in the final table.

Problem 4.1 6

QUESTION AS PRINTED 2080 Bhadra · Q2 (b) · [3+6]

When do we use classifier? You have the following information about the flower. Your job is to classify the given flower with SepalLength = 7, SepalWidth = 3.2, PetalLength = 4.7, and PetalWidth = 1.4. Use KNN algorithm for $K = 3$ with Euclidean distance matrix.

Id	Sepal_Lenth	Spal_Width	Petal_Lenth	Petal_Width	Labal
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	6	2.2	4	1	versicolor
5	6.1	2.9	4.7	1.4	versicolor
6	5.6	2.9	3.6	1.3	versicolor
7	6.7	3.1	4.4	1.4	versicolor

Column names are

the paper's own spellings.

Id	Tuple	Squared differences	d	Label
1	(5.1, 3.5, 1.4, 0.2)	$3.61 + 0.09 + 10.89 + 1.44 = 16.03$	4.004	setosa
2	(4.9, 3.0, 1.4, 0.2)	$4.41 + 0.04 + 10.89 + 1.44 = 16.78$	4.096	setosa
3	(4.7, 3.2, 1.3, 0.2)	$5.29 + 0 + 11.56 + 1.44 = 18.29$	4.277	setosa
4	(6.0, 2.2, 4.0, 1.0)	$1 + 1 + 0.49 + 0.16 = 2.65$	1.628	versicolor
5	(6.1, 2.9, 4.7, 1.4)	$0.81 + 0.09 + 0 + 0 = 0.90$	0.949	versicolor
6	(5.6, 2.9, 3.6, 1.3)	$1.96 + 0.09 + 1.21 + 0.01 = 3.27$	1.808	versicolor
7	(6.7, 3.1, 4.4, 1.4)	$0.09 + 0.01 + 0.09 + 0 = 0.19$	0.436	versicolor

Step: pick the 3 nearest

- Sorted: $d_7 = 0.436 < d_5 = 0.949 < d_4 = 1.628 < d_6 = 1.808 < d_1 = 4.004 \dots$
- The 3 nearest are **Id 7, Id 5, Id 4** → versicolor, versicolor, versicolor.

Final answer

versicolor (unanimous, 3 of 3).

- Note all 3 setosa tuples sit at distance > 4 : they have tiny petals, so they are nowhere near the query. The vote was never in doubt.

M5. Confusion matrix metrics (8 papers — highest-yield numerical in the syllabus)

DO THIS FIRST, EVERY TIME. Copy the row and column headings onto your answer sheet and mark which axis is **actual**. The IOE papers **flip the orientation between years**.

Proof: 75 Ash and 72 Ch print the *identical* four numbers 21, 6, 7, 41 — but 75 Ash has actual on the rows and 72 Ch has predicted on the rows. The result is that **sensitivity and precision swap places** (0.778 / 0.750 becomes 0.750 / 0.778). Same numbers, different answers. verified/added

Method

- Identify TP , FN , FP , TN from the orientation. Then $P = TP + FN$ (all actual positives), $N = FP + TN$ (all actual negatives).
- Accuracy = $\frac{TP+TN}{P+N}$ Error = 1–accuracy
- Sensitivity = recall = $TPR = \frac{TP}{P}$ Specificity = $TNR = \frac{TN}{N}$
- FPR = false alarm rate = $\frac{FP}{N} = 1 - \text{specificity}$ Precision = $\frac{TP}{TP+FP}$

Problem 5.1 5

QUESTION AS PRINTED 2074 Chaitra · Q3 · [2+3+5]

Draw clear block diagram depicting different stages in classification. Explain the inverse relation between precision and recall. Given the confusion matrix, determine accuracy, sensitivity and precision of the classifier model.

Actual \ Predicted	Positive	Negative
Positive	142	40
Negative	98	720

"Positive" is the paper's own spelling.

- $TP = 142, FN = 40, FP = 98, TN = 720 \rightarrow P = 182, N = 818, \text{total} = 1000$
- Accuracy = $\frac{142+720}{1000} = \mathbf{0.862}$
- Sensitivity = $\frac{142}{182} = \mathbf{0.780}$
- Precision = $\frac{142}{142+98} = \frac{142}{240} = \mathbf{0.592}$
- Specificity = $\frac{720}{818} = \mathbf{0.880}$

Problem 5.2 4

QUESTION AS PRINTED 2078 Bhadra · Q6 · [4]

Calculate: Accuracy, TPR, FPR and Precision for the given confusion matrix for a classifier.

		Actual Class	
		Class 1	Class 2
Predicted Class	Class 1	142	40
	Class 2	98	720

Same four numbers as 5.1, but the rows are Predicted here.

- Rows are predicted $\rightarrow TP = 142, FP = 40, FN = 98, TN = 720$
- $P = 142 + 98 = 240, N = 40 + 720 = 760$
- Accuracy = $\frac{142+720}{1000} = \mathbf{0.862}$ (unchanged – the diagonal is the same)
- TPR = $\frac{142}{240} = \mathbf{0.592}$ FPR = $\frac{40}{760} = \mathbf{0.053}$
- Precision = $\frac{142}{142+40} = \frac{142}{182} = \mathbf{0.780}$
- **Compare with 5.1:** sensitivity and precision have exchanged values. This is the orientation trap, live.

Problems 5.3 and 5.4 10

QUESTION AS PRINTED 2075 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	21	6
	Class 2	7	41

Calculate: Accuracy, Sensitivity, Specificity and Precision.

QUESTION AS PRINTED 2072 Chaitra · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		actual class	
		class1	class2
predicted class	class1	21	6
	class2	7	41

Calculate a. accuracy b. sensitivity c. specificity d. precision e. recall

Same four numbers, opposite orientation – that is the whole point of solving them side by side.

75 Ash – rows are Actual

- $TP = 21, FN = 6, FP = 7, TN = 41; P = 27, N = 48, \text{total} = 75$
- Accuracy = $\frac{62}{75} = \mathbf{0.827}$
- Sensitivity = $\frac{21}{27} = \mathbf{0.778}$
- Specificity = $\frac{41}{48} = \mathbf{0.854}$
- Precision = $\frac{21}{28} = \mathbf{0.750}$

72 Ch – rows are Predicted

- $TP = 21, FP = 6, FN = 7, TN = 41; P = 28, N = 47, \text{total} = 75$
- Accuracy = $\frac{62}{75} = \mathbf{0.827}$
- Sensitivity = recall = $\frac{21}{28} = \mathbf{0.750}$
- Specificity = $\frac{41}{47} = \mathbf{0.872}$
- Precision = $\frac{21}{27} = \mathbf{0.778}$

Problem 5.5 10

QUESTION AS PRINTED 2074 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	25	9
	Class 2	4	31

Calculate: a) Accuracy b) Sensitivity c) Specificity d) Precision

- $TP = 25, FN = 9, FP = 4, TN = 31; P = 34, N = 35$, total 69
- Accuracy = $\frac{56}{69} = \mathbf{0.812}$ Sensitivity = $\frac{25}{34} = \mathbf{0.735}$
- Specificity = $\frac{31}{35} = \mathbf{0.886}$ Precision = $\frac{25}{29} = \mathbf{0.862}$

Problem 5.6 3+5

QUESTION AS PRINTED 2080 Baishakh · Q3 · [3+5]

In what cases you cannot use Accuracy for performance measure, give some examples. Assume that you have the following confusion matrix. Calculate the Classification error, Sensitivity, False alarm rate Specificity.

		Actual Values	
		True	False
Predicated Values	True	1050	250
	False	150	950

- $TP = 1050, FP = 250, FN = 150, TN = 950; P = 1200, N = 1200$, total 2400
- Classification error = $\frac{250+150}{2400} = \mathbf{0.167}$
- Sensitivity = $\frac{1050}{1200} = \mathbf{0.875}$
- False alarm rate (FPR) = $\frac{250}{1200} = \mathbf{0.208}$
- Specificity = $\frac{950}{1200} = \mathbf{0.792}$ (check: $0.208 + 0.792 = 1 \checkmark$)

Problem 5.7 3+6

QUESTION AS PRINTED 2073 Shrawan · Q4 · [1+3+6]

Explain ROC. Using the following data, calculate TPR, FPR, precision for given confusion matrix.

	A	B
A	20	5
B	10	40

Classify, A = Yes, B = No

- $TP = 20, FN = 5, FP = 10, TN = 40; P = 25, N = 50$
- TPR = $\frac{20}{25} = \mathbf{0.80}$ FPR = $\frac{10}{50} = \mathbf{0.20}$
- Precision = $\frac{20}{30} = \mathbf{0.667}$
- The ROC point for this classifier is (0.20, 0.80), comfortably above the diagonal.

Problem 5.8 10

QUESTION AS PRINTED 2070 Chaitra · Q3 · [10]

An input sequence “A A B B A A B B” was used for classification. The Classifier ‘X’ predicted the sequences as: “A A B B A A B B” where as the Classifier ‘Y’ predicted the sequences as: “A A A B B A A B”. Develop the corresponding confusion matrix for the classifiers and find their corresponding.

- i) Accuracy ii) Precision iii) True Positive Rate iv) False Positive Rate

Step 1: tabulate Y position by position

(take A as the positive class)

Position	1	2	3	4	5	6	7	8	9	10
Actual	A	A	B	B	B	A	A	A	B	B
Y	A	A	A	A	B	B	A	A	A	B
Category	TP	TP	FP	FP	TN	FN	TP	TP	FP	TN

Classifier X

(identical to actual)

- $TP = 5, TN = 5, FP = 0, FN = 0$
- Accuracy = **1.0** Precision = **1.0**
- TPR = **1.0** FPR = **0**
- **Perfect classifier** – the ROC point is the top-left corner (0, 1).
- **Reading:** Y has decent recall (0.80) but poor precision (0.571) – it over-predicts A, firing 3 false alarms out of 5 actual Bs. Its ROC point (0.60, 0.80) is only just above the diagonal, so it is barely better than guessing.

Classifier Y

- $TP = 4, FN = 1, FP = 3, TN = 2; P = 5, N = 5$ (total 10 ✓)
- Accuracy = $\frac{4+2}{10} = \mathbf{0.60}$
- Precision = $\frac{4}{7} = \mathbf{0.571}$
- TPR = $\frac{4}{5} = \mathbf{0.80}$
- FPR = $\frac{3}{5} = \mathbf{0.60}$

M6. ANN gradients (backpropagation)**Method**

- Sigmoid and its derivative: $\sigma(z) = \frac{1}{1+e^{-z}}, \quad \sigma'(z) = \sigma(z)(1 - \sigma(z))$.
- Loss $E = \frac{1}{2}(o - t)^2 \rightarrow \frac{\partial E}{\partial o} = (o - t)$.
- Apply the **chain rule** backwards from the loss to the weight in question.

Problem 6.1 2+3+4

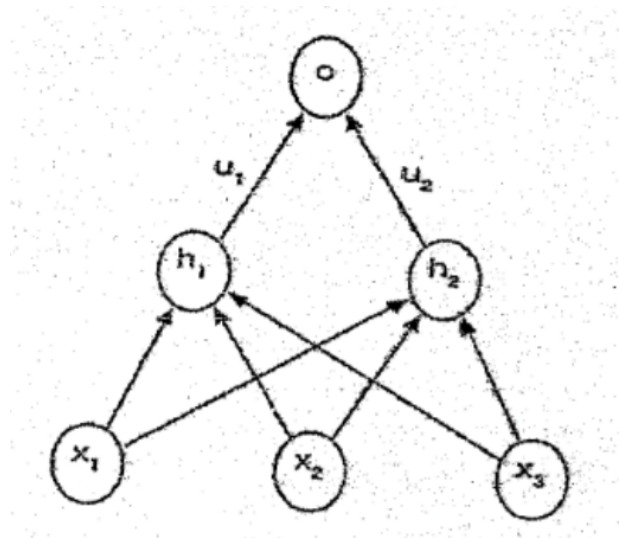
QUESTION AS PRINTED 2076 Chaitra · Q6 · [2+3+4]

Consider the multi-layer feed-forward neural network shown in the following figure. This neural network has three inputs x_1, x_2 and x_3 connected to a hidden layer consisting of two nodes h_1 and h_2 . The weight of the edge connecting x_i to h_j is w_{ji} . The two hidden nodes are connected to the output node o . The weight of the edge connecting the hidden node h_i to the output node o is u_i . The activation functions at hidden and output layers is set to sigmoid function defined as follows:

$$\sigma(\theta) = \frac{1}{1 + e^{-\theta}}$$

Using the target output t , the squared error is used as the loss function at the output node, and is defined as:

$$E(o, t) = \frac{1}{2}(o - t)^2$$



- Using the symbols given above, compute the activation at h_1 .
- Compute the gradient of the loss with respect to the output o .
- Compute the gradient of the loss with respect to the weight w_{12} .

(a) Activation at h_1

$$h_1 = \sigma(w_{11}x_1 + w_{12}x_2 + w_{13}x_3) = \frac{1}{1 + e^{-(w_{11}x_1 + w_{12}x_2 + w_{13}x_3)}}$$

(b) Gradient of the loss w.r.t. the output o

$$E(o, t) = \frac{1}{2}(o - t)^2 \Rightarrow \frac{\partial E}{\partial o} = (o - t)$$

(c) Gradient of the loss w.r.t. w_{12}

- w_{12} is the weight from x_2 into h_1 , so the chain runs $E \rightarrow o \rightarrow h_1 \rightarrow w_{12}$.
- Write $net_o = u_1h_1 + u_2h_2$ and $net_{h_1} = w_{11}x_1 + w_{12}x_2 + w_{13}x_3$.

$$\frac{\partial E}{\partial w_{12}} = \underbrace{\frac{\partial E}{\partial o}}_{(o-t)} \cdot \underbrace{\frac{\partial o}{\partial net_o}}_{o(1-o)} \cdot \underbrace{\frac{\partial net_o}{\partial h_1}}_{u_1} \cdot \underbrace{\frac{\partial h_1}{\partial net_{h_1}}}_{h_1(1-h_1)} \cdot \underbrace{\frac{\partial net_{h_1}}{\partial w_{12}}}_{x_2}$$

$$\frac{\partial E}{\partial w_{12}} = (o - t) o(1 - o) u_1 h_1(1 - h_1) x_2$$

- Sanity checks:** the factor x_2 must appear (no input, no gradient); u_1 appears and u_2 does not, because w_{12} only influences o through h_1 .
- Weight update:** $w_{12} \leftarrow w_{12} - l \frac{\partial E}{\partial w_{12}}$.

Ch 4 – Association Analysis – Numerical Problems & Solutions

5 methods · 14 PYQ instances · every itemset count recomputed from the printed transaction tables. **This is the heaviest numerical chapter in the syllabus.**

Methods

- **M1** Apriori: candidate generation, pruning, frequent itemsets
- **M2** Rule generation from frequent itemsets
- **M3** FP-tree construction and FP-Growth mining
- **M4** Pattern evaluation: lift and the correlation measures
- **M5** Subsequence enumeration

Convert the threshold to a COUNT before you start. “min support 50%” on 6 transactions means a count of 3. Nearly every lost mark in this chapter comes from mixing percentages and counts half way through. Write $\text{min_sup_count} = N$ at the top and use only that.

Also: **support is over ALL transactions; confidence is over the antecedent's transactions only.**

M1. Apriori

Method

1. Scan once, count items $\rightarrow$ keep those $\geq \text{min_sup_count} \rightarrow L_1$.
 2. **Join** L_{k-1} with itself: combine 2 itemsets only if their first $k-2$ items match.
 3. **Prune**: discard a candidate if *any* of its $(k-1)$ -subsets is missing from L_{k-1} .
 4. **Count** survivors against D , keep those meeting the threshold $\rightarrow L_k$.
 5. Stop when L_k is empty.
- **Show the prune step explicitly** – it is where the method marks are.

Problem 1.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q3 (b) · [2+7]

What is Association Analysis? Explain with different use cases? Use the Apriori algorithm to find the frequent itemsets. Assume minimum support count is 4 and confidence is 80%.

TID	Items
T100	F, A, C, D, G, I, M, P, N
T101	A, B, C, D, F, L, M, O, P
T102	B, F, H, V, J, O, P
T103	B, C, K, S, A, V
T104	L, A, F, C, E, P, M, N, V
T105	I, B, A, P, S, M
T106	F, A, C, I, B, A, P

T106 genuinely prints “F, A, C, I, B, A, P” – **A is repeated; a transaction is a set**, so treat it as {F,A,C,I,B,P}.

Step 1: item counts

(7 transactions)

- A:6 · P:6 · B:5 · C:5 · F:5 · M:4 · I:3 · V:3 · D:2 · L:2 · N:2 · O:2 · S:2 · others 1
- $L_1 = \{\mathbf{A(6)}, \mathbf{B(5)}, \mathbf{C(5)}, \mathbf{F(5)}, \mathbf{M(4)}, \mathbf{P(6)}\}$

Step 2: L_2

(15 candidate pairs; counts shown, ≥ 4 kept)

- Kept: **AB(4), AC(5), AF(4), AM(4), AP(5), BP(4), CF(4), CP(4), FP(5), MP(4)**
- Dropped: BC(3), BF(3), BM(2), CM(3), FM(3)

Step 3: L_3

- ABP: all 3 subsets present $\rightarrow$ counted = $3 \times$
- **ACF(4), ACP(4), AFP(4), AMP(4), CFP(4) ✓**

- Pruned without counting: ABC ($BC \notin L_2$), ABF (BF), ABM (BM), ACM (CM), AFM (FM)

Step 4: L_4

- ACFP: subsets ACF ✓ ACP ✓ AFP ✓ CFP ✓ → candidate. Count = 4 (T100, T101, T104, T106) ✓
- $L_4 = \{\mathbf{ACFP(4)}\}$. L_5 empty → stop.

Final answer

Maximal frequent itemset = **{A, C, F, P} with support 4/7 = 57%**.

- **Check:** all $2^4 - 2 = 14$ proper subsets of ACFP do appear in $L_1 - L_3$ above. ✓

Problem 1.2 5

QUESTION AS PRINTED 2080 Baishakh · Q5 · [4+5]

Generally, we will be more interested in associated rules with high confidence. However, often we will not be interested in association rules that have a confidence of 100%. Why? Then specifically explain why association rules with 99% confidence may be interesting (i.e., what might they indicate)? Identify the candidate and large item sets of the following transaction table using Apriori algorithm with minimum support 2.

TID	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

Step 1: L_1 , count every item across the 4 transactions

- A appears in T1, T3 → 2 B appears in T2, T3, T4 → 3
- C appears in T1, T2, T3 → 3 D appears in T1 only → 1 (**dropped**)
- E appears in T2, T3, T4 → 3
- $L_1 = \{\mathbf{A(2)}, \mathbf{B(3)}, \mathbf{C(3)}, \mathbf{E(3)}\}$

Step 2: C_2 , join L_1 with itself, then count each pair against the transactions

- AB: only T3 → 1 × AC: T1, T3 → 2 ✓ AE: only T3 → 1 ×
- BC: T2, T3 → 2 ✓ BE: T2, T3, T4 → 3 ✓ CE: T2, T3 → 2 ✓
- $L_2 = \{\mathbf{AC(2)}, \mathbf{BC(2)}, \mathbf{BE(3)}, \mathbf{CE(2)}\}$; dropped AB(1), AE(1)

Step 3: C_3 , join L_2 with itself, prune, then count

- BC and BE share prefix B → candidate **BCE**. Subsets BC ✓, BE ✓, CE ✓ all in L_2 → not pruned.
- AC joins with nothing else in L_2 (no other pair starts with A) → no candidate.
- Count BCE against the transactions: T2 = {B, C, E} ✓, T3 = {A, B, C, E} ✓ → count = 2 ✓ (meets threshold).
- $L_3 = \{\mathbf{BCE(2)}\}$. C_4 is empty (only one 3-itemset survived) → stop.

Final answer

Frequent itemsets: A, B, C, E, AC, BC, BE, CE, BCE.

Problem 1.3 10

QUESTION AS PRINTED 2070 Chaitra · Q4 · [10]

Explain Apriori algorithm. Use Apriori to generate frequent item sets with support of 50% for the following transaction database.

TID	Items
1	ACD
2	BD
3	ABCE
4	BDF

4 transactions, 50% $\rightarrow$ min count = 2.

- L_1 : A(2), B(3), C(2), D(3) (E:1, F:1 dropped)
- C_2 : AB(1) $\times$ AC(2) $\checkmark$ AD(1) $\times$ BC(1) $\times$ BD(2) $\checkmark$ CD(1) $\times$
- $L_2 = \{\mathbf{AC(2)}, \mathbf{BD(2)}\}$
- C_3 : AC and BD share no common prefix $\rightarrow$ **no candidates**. Stop.

Final answer

Frequent itemsets: A, B, C, D, AC, BD.

Problems 1.4 and 1.5 5+3 8

QUESTION AS PRINTED 2076 Chaitra · Q7 · [5+3]

Consider the transaction data shown in the following table from a fast food restaurant.

Meal Item	List of Item IDs
Order:1	M1, M2, M5
Order:2	M2, M4
Order:3	M2, M3
Order:4	M1, M2, M4
Order:5	M1, M3
Order:6	M2, M3
Order:7	M1, M3
Order:8	M1, M2, M3, M5
Order:9	M1, M2, M3

There are 9 distinct transactions (Order: 1 – Order: 9) and each transaction involves between 2 and 4 meal items. There are a total of 5 meal items that are involved in the transactions. For simplicity, the meal items have been assigned short names (M1–M5). Assume that the minimum support is 2/9 and the minimum confidence is 7/9.

- Apply the Apriori algorithm to the dataset of transactions and identify all frequent k -itemsets.
- Find all strong association rules of the form: $X \wedge Y \rightarrow Z$ and note their confidence values.

QUESTION AS PRINTED 2074 Ashwin · Q5 · [8]

Identify the candidate, frequent item sets and association rules for the following transaction data using Apriori algorithm.

TID	ITEMS
1	M1, M2, M5
2	M2, M4
3	M2, M3
4	M1, M2, M4
5	M1, M3
6	M2, M3
7	M1, M3
8	M1, M2, M3, M5
9	M1, M2, M3

Take minimum support = 20%, minimum confidence 80%

Identical 9-transaction table, and both thresholds work out to a count of 2, so the frequent itemsets are the same; only the rule filter differs.

Frequent itemsets

(9 transactions, min count 2)

- **Step 1**, L_1 : M1 in T1,T4,T5,T7,T8,T9 $\rightarrow$ 6. M2 in T1,T2,T3,T4,T6,T8,T9 $\rightarrow$ 7. M3 in T3,T5,T6,T7,T8,T9 $\rightarrow$ 6. M4 in T2,T4 $\rightarrow$ 2. M5 in T1,T8 $\rightarrow$ 2.

- $L_1 = \{M1(6), M2(7), M3(6), M4(2), M5(2)\}$ (nothing dropped, all 5 items clear count 2)
- **Step 2**, $C_2 \rightarrow L_2$: check each pair against the table. M1M2: T1,T4,T8,T9 $\rightarrow 4 \checkmark$. M1M3: T5,T7,T8,T9 $\rightarrow 4 \checkmark$. M1M4: T4 only $\rightarrow 1 \times$. M1M5: T1,T8 $\rightarrow 2 \checkmark$. M2M3: T3,T6,T8,T9 $\rightarrow 4 \checkmark$. M2M4: T2,T4 $\rightarrow 2 \checkmark$. M2M5: T1,T8 $\rightarrow 2 \checkmark$. M3M4: never together $\rightarrow 0 \times$. M3M5: T8 only $\rightarrow 1 \times$. M4M5: never together $\rightarrow 0 \times$.
- $L_2 = \{M1M2(4), M1M3(4), M1M5(2), M2M3(4), M2M4(2), M2M5(2)\}$; dropped M1M4(1), M3M4(0), M3M5(1), M4M5(0)
- **Step 3**, C_3 : join pairs sharing a prefix, prune by subset, then count. M1M2M3 (from M1M2 + M1M3, needs M2M3 too $\rightarrow$ present) $\rightarrow$ count in T8,T9 $\rightarrow 2 \checkmark$. M1M2M5 (from M1M2 + M1M5, needs M2M5 $\rightarrow$ present) $\rightarrow$ count in T1,T8 $\rightarrow 2 \checkmark$. M1M2M4 pruned (subset M1M4 $\notin L_2$). M2M3M5 pruned (subset M3M5 $\notin L_2$). M1M3M5 pruned (subset M3M5 $\notin L_2$).
- $L_3 = \{M1M2M3(2), M1M2M5(2)\}$
- **Step 4**, C_4 : only candidate is M1M2M3M5 (from M1M2M3 + M1M2M5), pruned because its subset M1M3M5 $\notin L_3 \rightarrow L_4$ empty $\rightarrow$ stop.

1.4 – rules $X \wedge Y \Rightarrow Z$, min conf $7/9 = 0.778$

- From M1M2M3 (sup 2): all three give $2/4 = 0.5 \times$
- From M1M2M5 (sup 2):
 - $M1 \wedge M2 \Rightarrow M5 = 2/4 = 0.5 \times$
 - $M1 \wedge M5 \Rightarrow M2 = 2/2 = 1.0 \checkmark$
 - $M2 \wedge M5 \Rightarrow M1 = 2/2 = 1.0 \checkmark$

Answer

2 strong rules, both at 100% confidence.

1.5 – all rules, min conf 0.80

- $M5 \Rightarrow M1 = 2/2 = 1.0 \checkmark$ $M5 \Rightarrow M2 = 2/2 = 1.0 \checkmark$
- $M4 \Rightarrow M2 = 2/2 = 1.0 \checkmark$
- $M5 \Rightarrow M1 \wedge M2 = 2/2 = 1.0 \checkmark$
- Plus the two 3-itemset rules from 1.4.
- All rules with M1, M2 or M3 as antecedent fail (best is $4/6 = 0.667$).
- **Pattern**: only the *rare* items (M4, M5) produce confident rules – they never appear without their partners.

Problem 1.6 8

QUESTION AS PRINTED 2079 Bhadra · Q5 · [8]

Derive association rule for the following market basket transactions.

Minimum support = 50% Minimum confidence = 80%

Transaction ID	Item Set
1	A,B
2	A,D
3	A,C
4	B,E
5	B,D,E
6	A,E,C

This one has a surprise.

- 6 transactions, 50% $\rightarrow$ min count = 3.
- L_1 : A(4), B(3), E(3) (C:2 $\times$, D:2 $\times$)
- C_2 : AB = 1 $\times$ AE = 1 $\times$ BE = 2 $\times$
- $L_2 = \emptyset$

Final answer

- **No frequent 2-itemset exists, so no association rule can be derived at 50% support.** Say this explicitly – it is the correct answer, not a dead end.
- **Show you understand why**: the data is spread thin, every pair occurs at most twice.
- **If min support is relaxed to 33.3%** (count 2): $L_2 = \{AC(2), BE(2)\}$, and the only rule clearing 80% confidence is $C \Rightarrow A$ at $2/2 = 100\%$. Worth adding as a one-line remark. verified/added

M2. Rule generation from frequent itemsets

Method

- For each frequent itemset l with $|l| \geq 2$, take every non-empty proper subset s :

$$s \Rightarrow (l - s) \quad \text{with} \quad \text{conf} = \frac{\text{sup_count}(l)}{\text{sup_count}(s)}$$

- Keep it if $\text{conf} \geq \text{min_conf}$. A k -itemset gives $2^k - 2$ candidate rules.

- **The support of every rule from l is the same** ($= \text{sup}(l)$); only confidence varies.
- **Shortcut:** if $s \Rightarrow (l - s)$ fails, every rule whose antecedent is a *subset* of s also fails. Prune downward.

Worked example

PROBLEM not a PYQ · the rule half of 2081 Baishakh Q3 (b) verified/added

The frequent itemset $l = \{K, E, O\}$ has support count 3 in the 5-transaction database of Problem 3.1, where $\text{sup}(K) = 5$, $\text{sup}(E) = 4$, $\text{sup}(O) = 3$, $\text{sup}(K, E) = 4$, $\text{sup}(K, O) = 3$ and $\text{sup}(E, O) = 3$. List every association rule that l can generate and its confidence, and say which are strong at a minimum confidence of 80%.

From $l = \{K, E, O\}$ with $\text{sup} = 3$ (see Problem 3.1), $|l| = 3 \rightarrow 6$ candidate rules:

Rule	Confidence	Value	$\geq 80\%$?
$K \Rightarrow E, O$	$3/5$	0.60	×
$E \Rightarrow K, O$	$3/4$	0.75	×
$O \Rightarrow K, E$	$3/3$	1.00	✓
$K, E \Rightarrow O$	$3/4$	0.75	×
$K, O \Rightarrow E$	$3/3$	1.00	✓
$E, O \Rightarrow K$	$3/3$	1.00	✓

- Every rule with **O** in the antecedent succeeds, because O never appears without K and E.

M3. FP-tree construction and FP-Growth mining

Method

1. **Scan 1:** count items, drop infrequent ones, sort the rest by descending support $\rightarrow$ **F-list**. **State your tie-break rule.**
2. **Scan 2:** rewrite each transaction in F-list order, insert as a path, sharing prefixes and incrementing counts. Build the header table with node-links.
3. **Mine bottom-up** (least frequent item first): conditional pattern base $\rightarrow$ conditional FP-tree $\rightarrow$ recurse.

Problem 3.1 2+7

QUESTION AS PRINTED 2081 Baishakh · Q3 (b) · [2+7]

What is limitation of Apriori algorithm compared to FP-growth? A database has 5 transactions, given in table below. Let min support = 60% and min confidence = 80%.

TID	Item_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

- i) Find all frequent itemsets using FP-growth.
 - ii) List all of the strong association rules (with support and confidence).
- **T500 prints O twice** — a transaction is a set, so it is {C,O,K,I,E}.
 - 5 transactions, 60% $\rightarrow$ **min count = 3**.

Step 1: counts and F-list

- K:5, E:4, M:3, O:3, Y:3 ✓ N:2, C:2, D:1, A:1, U:1, I:1 ×
- **F-list = K(5), E(4), M(3), O(3), Y(3)** (ties at 3 broken alphabetically)

Step 2: reordered transactions

- T100 → K, E, M, O, Y
- T200 → K, E, O, Y
- T300 → K, E, M
- T400 → K, M, Y
- T500 → K, E, O

Step 3: the FP-tree

- root
- → K:5
- → E:4
- → M:2 → O:1 → Y:1
- → O:2 → Y:1
- → M:1 → Y:1
- **Check:** counts on each level sum correctly – K:5 = E:4 + M:1. ✓

All frequent itemsets

- 1-itemsets: K:5, E:4, M:3, O:3, Y:3
- 2-itemsets: {K,E}:4, {K,M}:3, {K,O}:3, {E,O}:3, {K,Y}:3
- 3-itemsets: {K,E,O}:3

Strong rules (min conf 80%)

- $K \Rightarrow E = 4/5 = 0.80$ ✓ $E \Rightarrow K = 4/4 = 1.0$ ✓
- $M \Rightarrow K = 3/3 = 1.0$ ✓ $O \Rightarrow K = 3/3 = 1.0$ ✓ $Y \Rightarrow K = 3/3 = 1.0$ ✓ $O \Rightarrow E = 3/3 = 1.0$ ✓
- $K, O \Rightarrow E = 1.0$ ✓ $E, O \Rightarrow K = 1.0$ ✓ $O \Rightarrow K, E = 1.0$ ✓
- Failing: $K \Rightarrow M$, $K \Rightarrow O$, $K \Rightarrow Y$ (all $3/5 = 0.6$), $E \Rightarrow O$ (0.75), $K, E \Rightarrow O$ (0.75), $E \Rightarrow K, O$ (0.75).

Problem 3.2 4+4

QUESTION AS PRINTED 2075 Chaitra · Q4 · [4+4]

What do you mean by frequent Pattern growth, draw FP-tree with given tabular data.

TID	Items
01	f, a, c, d, g, i, m, p
02	a, b, c, f, l, m, o
03	b, f, h, j, o, w
04	b, c, k, s, p
05	A, f, c, e, l, p, m, n

Min support is not printed; use 3, which is the standard for this dataset. verified/added**Counts and F-list**

- f:4, c:4, a:3, b:3, m:3, p:3 ✓ l:2, o:2, and d,g,i,h,j,k,s,e,n,w all 1 ×
- **F-list = f, c, a, b, m, p**
- **T05 prints a capital "A"** – same item as a.

Reordered transactions

- 01 → f, c, a, m, p 02 → f, c, a, b, m
- 03 → f, b 04 → c, b, p
- 05 → f, c, a, m, p

Step 4: mine bottom-up

Item	Conditional pattern base	Frequent patterns
Y	{K,E,M:1}, {K,E:1}, {K,M:1}	{K,Y}:3
O	counts K:3, E:2, M:2 {K,E,M:1}, {K,E:2}	{K,O}:3, {E,O}:3,
M	counts K:3, E:3 {K,E:2}, {K:1}	{K,E,O}:3 {K,M}:3
E	counts K:3, E:2 {K:4}	{K,E}:4
K	–	{K}:5

The FP-tree

- root
- → f:4
- → c:3 → a:3
- → m:2 → p:2
- → b:1 → m:1
- → b:1
- → c:1 → b:1 → p:1
- **Note the second root child:** transaction 04 starts with c, not f, so it cannot share the main prefix.

Problem 3.3 4+4

QUESTION AS PRINTED 2078 Bhadra · Q5 · [4+4]

Consider the given transactional database from a grocery store. Use a support threshold of 33.34% and confidence threshold of 60% to compute the following:

- Build a frequent pattern tree (FP-Tree). Show for each transaction how the tree evolves.
- Use FP-Growth algorithm to discover the frequent itemsets from this FP-tree.

Transaction ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

- 6 transactions, 33.34% → **min count = 2**.
- Counts: HotDogs 4, Chips 4, Coke 3, Buns 2, Ketchup 2 — **all frequent**.
- F-list = Chips(4), HotDogs(4), Coke(3), Buns(2), Ketchup(2)** (4-way tie broken alphabetically).

Tree, built transaction by transaction

- T1 (HotDogs, Buns, Ketchup): root → H:1 → Bu:1 → Ke:1
- T2 (HotDogs, Buns): H:2, Bu:2
- T3 (Chips, HotDogs, Coke): new branch
Ch:1 → H:1 → Co:1
- T4 (Chips, Coke): Ch:2, new child Co:1
- T5 (Chips, Ketchup): Ch:3, new child Ke:1
- T6 (Chips, HotDogs, Coke): Ch:4, H:2, Co:2

Final tree

- root
- Chips:4 → {HotDogs:2 → Coke:2, Coke:1, Ketchup:1}
- HotDogs:2 → Buns:2 → Ketchup:1

Mining

Item	Cond. pattern base	Patterns
Ketchup	{H,Bu:1}, {Ch:1}	none (all counts 1)
Buns	{H:2}	{H,Bu:2}
Coke	{Ch,H:2}, {Ch:1}	{Ch,Co:3}, {H,Co:2}, {Ch,H,Co:2}
HotDogs	{Ch:2}	{Ch,H:2}
Chips	—	{Ch:4}

Frequent itemsets

- Chips 4, HotDogs 4, Coke 3, Buns 2, Ketchup 2
- {Chips,Coke} 3, {Chips,HotDogs} 2, {HotDogs,Coke} 2, {HotDogs,Buns} 2
- {Chips,HotDogs,Coke} 2

Problem 3.4 2+8

QUESTION AS PRINTED 2074 Chaitra · Q6 · [2+8]

How does FP growth approach overcomes the disadvantages of Apriori algorithm. For the transaction data given in table generate FP-Tree.

Transaction ID	Item set
T1	Camera, Laptop, Pen drive
T2	Laptop, Pen drive
T3	Laptop, Mobile, Earphone
T4	Earphone, Mobile
T5	Camera, Earphone
T6	Laptop, Mobile, Earphone

Min support is not printed; use 2.

- Counts: Laptop 4, Earphone 4, Mobile 3, Camera 2, Pen drive 2 — **all frequent**.
- F-list = Earphone(4), Laptop(4), Mobile(3), Camera(2), Pen drive(2)**
- Reordered: T1 (L, Ca, Pd) · T2 (L, Pd) · T3 (E, L, Mo) · T4 (E, Mo) · T5 (E, Ca) · T6 (E, L, Mo)
- Tree:** root → Earphone:4 → {Laptop:2 → Mobile:2, Mobile:1, Camera:1}; and root → Laptop:2 → {Camera:1 → PenDrive:1, PenDrive:1}
- Key frequent pattern: **{Earphone, Laptop, Mobile}:2**

M4. Pattern evaluation: lift and correlation

Method

$$\text{lift}(A, B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{\text{conf}(A \Rightarrow B)}{\text{sup}(B)} \quad \chi^2 = \sum \frac{(\text{obs} - \text{exp})^2}{\text{exp}}$$

- $\text{lift} > 1$ positive · $= 1$ independent · < 1 negative.

Problem 4.1 8

QUESTION AS PRINTED 2070 Chaitra · Q5 · [8]

Why is pattern evaluation important in association rule mining? Explain with example the statistical based measures used for measuring interestingness of association rules.

The paper prints no data — the example is yours to supply. The standard *game* / *video* contingency table below is the one every textbook uses. verified/added

Given

10,000 transactions: 6,000 contain *game*, 7,500 contain *video*, 4,000 contain both.

Step 1: the support–confidence view says the rule is strong

- $\text{sup}(\text{game} \Rightarrow \text{video}) = 4000/10000 = 40\%$ (passes a 30% threshold)
- $\text{conf}(\text{game} \Rightarrow \text{video}) = 4000/6000 = 66.7\%$ (passes a 60% threshold)

Step 2: but look at the consequent alone

- $P(\text{video}) = 7500/10000 = 75\%$
- **Buying a game DROPS the chance of buying a video from 75% to 66.7%.** The rule is actively misleading.

Step 3: lift catches it

$$\text{lift} = \frac{0.40}{0.60 \times 0.75} = \frac{0.40}{0.45} = 0.89 < 1 \rightarrow \text{negatively correlated}$$

Step 4: χ^2 agrees

	game	<i>game</i>	row
video	4000 (4500)	3500 (3000)	7500
<i>video</i>	2000 (1500)	500 (1000)	2500
col	6000	4000	10,000

$$\chi^2 = \frac{500^2}{4500} + \frac{500^2}{3000} + \frac{500^2}{1500} + \frac{500^2}{1000} = 55.6 + 83.3 + 166.7 + 250 = 555.6$$

- Large $\chi^2 \rightarrow$ correlated; observed (4000) < expected (4500) $\rightarrow$ **negatively**.

Final answer

- **Pattern evaluation matters because support and confidence alone certify rules that are false.** Add a correlation measure before reporting any rule.
- Mention that lift and χ^2 are **not null-invariant**, so for sparse data prefer **all_conf**, **max_conf**, **Kulczynski** or **cosine**.

M5. Subsequence enumeration

Method

- A **k -subsequence** contains k items in total (not k elements).
- Choose how many items to take from each element, keeping element order; a chosen element must be non-empty to remain in the sequence.
- **Enumerate systematically** by the split (a, b, c, d) of items taken per element, then remove duplicates.

Problem 5.1 3

QUESTION AS PRINTED 2076 Chaitra · Q8 (a) · [3]

List all the 4-subsequences contained in the data sequence: $\langle \{1, 3\} \{2\} \{2, 3\} \{4\} \rangle$

Given

$e_1 = \{1, 3\}$, $e_2 = \{2\}$, $e_3 = \{2, 3\}$, $e_4 = \{4\}$; 6 items in total.

Step 1: enumerate the item-count splits

(a, b, c, d) with $a \leq 2, b \leq 1, c \leq 2, d \leq 1$, $\text{sum} = 4$

(a, b, c, d)	Ways	Subsequences
(2,1,1,0)	2	$\langle \{1, 3\} \{2\} \{2\} \rangle$, $\langle \{1, 3\} \{2\} \{3\} \rangle$
(2,1,0,1)	1	$\langle \{1, 3\} \{2\} \{4\} \rangle$
(2,0,2,0)	1	$\langle \{1, 3\} \{2, 3\} \rangle$
(2,0,1,1)	2	$\langle \{1, 3\} \{2\} \{4\} \rangle$ (duplicate), $\langle \{1, 3\} \{3\} \{4\} \rangle$
(1,1,2,0)	2	$\langle \{1\} \{2\} \{2, 3\} \rangle$, $\langle \{3\} \{2\} \{2, 3\} \rangle$
(1,1,1,1)	4	$\langle \{1\} \{2\} \{2\} \{4\} \rangle$, $\langle \{1\} \{2\} \{3\} \{4\} \rangle$, $\langle \{3\} \{2\} \{2\} \{4\} \rangle$, $\langle \{3\} \{2\} \{3\} \{4\} \rangle$
(1,0,2,1)	2	$\langle \{1\} \{2, 3\} \{4\} \rangle$, $\langle \{3\} \{2, 3\} \{4\} \rangle$
(0,1,2,1)	1	$\langle \{2\} \{2, 3\} \{4\} \rangle$

Final answer

- 15 selections, but $\langle \{1, 3\} \{2\} \{4\} \rangle$ arises twice (the $\{2\}$ can come from e_2 or e_3), so there are **14 distinct 4-subsequences**.
- **The exam usually only wants “at least four”** 79 Bh, so listing the first few from a clearly stated method is enough. **Show the method, not just the list.**

Problem 5.2 3

QUESTION AS PRINTED 2079 Bhadra · Q6 (a) · [3+3]

How do you handle the categorical attributes in data mining process? Explain with example. Generate the at least four subsequences from the given sequence: $\langle \{2, 3, 5\}, \{6, 7, 8\}, \{9, 1\}, \{7, 4\} \rangle$.

- Any order-preserving selection works. Four valid examples:
 - $\langle \{2\} \{6\} \rangle$ $\langle \{3, 5\} \{7\} \{9\} \rangle$
 - $\langle \{2, 3\} \{6, 7\} \{1\} \{4\} \rangle$ $\langle \{5\} \{8\} \{9, 1\} \{7\} \rangle$
- **Must hold:** element order preserved, and each chosen sub-element is a subset of the original element. **Must not do:** reorder elements, or merge items from two elements into one.

Ch 5 – Cluster Analysis – Numerical Problems & Solutions

3 methods · 11 PYQ instances · every distance recomputed. K-means appears in 7 papers and hierarchical dendrograms in 4, so these two methods alone are worth around 60 marks across the archive.

Methods

- **M1** K-means 78 Bh, 74 Ash, 75 Ash, 71 Shr, 76 Ch, 81 Ba
- **M2** Agglomerative hierarchical clustering + dendrogram 80 Bh, 80 Ba, 76 Ch
- **M3** DBSCAN point classification *practice*

M1. K-means

Method

1. State K and the initial centroids. **Always write down which points you seeded with** — the answer depends on it and the examiner needs to follow you.
 2. Build a distance table: every point against every centroid.
 3. Assign each point to its nearest centroid.
 4. Recompute each centroid as the **mean** of its members.
 5. Repeat until the assignment does not change. **That repetition is the convergence proof.**
- You may compare *squared* distances to avoid square roots — the ordering is identical. Say that you are doing it.

Problems 1.1 and 1.2 2+6 2+8

QUESTION AS PRINTED 2078 Bhadra · Q8 · [2+6]

Write K-means clustering algorithm. Generate two clusters from following dataset using K-means clustering.

Instance	A	B
1	1	2
2	2.5	1
3	3.5	1.5
4	4	1
5	3.5	2.5
6	5	3

QUESTION AS PRINTED 2074 Ashwin · Q7 · [2+8]

Write K-means algorithm and find clusters for following data set.

Instance	X	Y
1	1.0	2.0
2	2.5	1.0
3	3.5	1.5
4	4.0	1.0
5	3.5	2.5
6	5.0	3.0

(Take $K = 2$)

The same 6 instances, only the column names differ.

Seeds

Take the two **most widely separated** points, $c_1 = p_1(1, 2)$ and $c_2 = p_6(5, 3)$. (A standard, defensible heuristic; state it.)

Iteration 1

Point	d to $c_1(1, 2)$	d to $c_2(5, 3)$	Cluster
$p_1(1, 2)$	0	4.123	C1
$p_2(2.5, 1)$	1.803	3.202	C1
$p_3(3.5, 1.5)$	2.550	2.121	C2
$p_4(4, 1)$	3.162	2.236	C2
$p_5(3.5, 2.5)$	2.550	1.581	C2
$p_6(5, 3)$	4.123	0	C2

- $C1 = \{p_1, p_2\} \rightarrow$ new centroid $(\frac{1+2.5}{2}, \frac{2+1}{2}) = (1.75, 1.5)$
- $C2 = \{p_3, p_4, p_5, p_6\} \rightarrow$ new centroid $(\frac{3.5+4+3.5+5}{4}, \frac{1.5+1+2.5+3}{4}) = (4, 2)$

Iteration 2

Point	d to $(1.75, 1.5)$	d to $(4, 2)$	Cluster
p_1	0.901	3.000	C1
p_2	0.901	1.803	C1
p_3	1.750	0.707	C2
p_4	2.305	1.000	C2
p_5	2.016	0.707	C2
p_6	3.580	1.414	C2

- **Assignment unchanged $\rightarrow$ converged.**

Final answer

- **Cluster 1** = $\{(1, 2), (2.5, 1)\}$, centroid $(1.75, 1.5)$
- **Cluster 2** = $\{(3.5, 1.5), (4, 1), (3.5, 2.5), (5, 3)\}$, centroid $(4, 2)$
- Converged in **2 iterations**.

Seed sensitivity — worth a line, since the question asks for limitations. Seeding instead with the *first* two points, $c_1 = p_1(1, 2)$ and $c_2 = p_2(2.5, 1)$, gives $C1 = \{p_1\}$ and $C2 = \{p_2, \dots, p_6\}$, which also converges but leaves a **degenerate 1-point cluster**. Same data, same K , **different answer** — this is exactly the “sensitive to initial centroids / converges to a local optimum” limitation. verified/added

Problem 1.3 4+6

QUESTION AS PRINTED 2075 Ashwin · Q7 · [4+6]

Explain K-means clustering with limitation. Generate two clusters from following dataset using K-means clustering.

A	B
1	2
2.5	4.5
4	6
3.5	4
4	5.5
3	6

- Seeds: $c_1 = (1, 2)$ (clearly isolated), $c_2 = (4, 6)$.
- **Iteration 1:** $(1, 2) \rightarrow C1$. All of $(2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)$ are nearer $(4, 6) \rightarrow C2$.
 - Eg $(2.5, 4.5)$: d to $c_1 = \sqrt{1.5^2 + 2.5^2} = 2.92$; to $c_2 = \sqrt{1.5^2 + 1.5^2} = 2.12 \rightarrow C2$.
 - Eg $(3.5, 4)$: d to $c_1 = \sqrt{2.5^2 + 2^2} = 3.20$; to $c_2 = \sqrt{0.5^2 + 2^2} = 2.06 \rightarrow C2$.
- New centroids: $C1 = (1, 2)$; $C2 = (\frac{2.5+4+3.5+4+3}{5}, \frac{4.5+6+4+5.5+6}{5}) = (3.4, 5.2)$
- **Iteration 2:** every C2 member stays nearer $(3.4, 5.2)$ than $(1, 2) \rightarrow$ unchanged $\rightarrow$ **converged**.

Final answer

$C1 = \{(1, 2)\}$, $C2 = \{(2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)\}$, centroid $(3.4, 5.2)$.

- **Note:** $(1, 2)$ is a genuine outlier here, so the 1-point cluster is the honest answer, not a seeding artefact. Say so, and add that **K-medoids** or **DBSCAN** would treat it as noise.

Problem 1.4 10

QUESTION AS PRINTED 2071 Shrawan · Q8 · [10]

Explain K-means clustering with limitation Use k-means clustering to cluster the following dataset.

A	B
1.0	1.0
1.5	2.0
3.0	4.0
5.0	7.0
3.5	5.0
4.5	5.0
3.5	4.5

The paper prints no K ; take $K = 2$, the value its sister questions use.

- Seeds $c_1 = (1, 1)$, $c_2 = (5, 7)$.
- **Iteration 1:** $(1, 1)$ and $(1.5, 2) \rightarrow C1$; the rest $\rightarrow C2$.
 - $(3, 4)$: d to $c_1 = \sqrt{4+9} = 3.61$; to $c_2 = \sqrt{4+9} = 3.61$ – **an exact tie**. Break it toward $C2$ (state your rule).
- Centroids: $C1 = (1.25, 1.5)$; $C2 = \left(\frac{3+5+3.5+4.5+3.5}{5}, \frac{4+7+5+5+4.5}{5}\right) = (3.9, 5.1)$
- **Iteration 2:** assignments unchanged $\rightarrow$ converged.

Final answer

$C1 = \{(1, 1), (1.5, 2)\}$ centroid $(1.25, 1.5)$; $C2 = \{(3, 4), (5, 7), (3.5, 5), (4.5, 5), (3.5, 4.5)\}$ centroid $(3.9, 5.1)$.

Problem 1.5 8

QUESTION AS PRINTED 2076 Chaitra · Q9 · [8]

Given the matrix X whose rows represent different data points, perform a k-means clustering on this dataset using the Euclidean distance as the distance function. Here K is chosen as 3. The center of the 3 clusters are initialized as red $(6.2, 3.2)$, green $(6.6, 3.7)$ and blue $(6.5, 3.0)$. Provide the final cluster centers and comment on the number of iterations required for the clusters to converge.

$$X = \begin{bmatrix} 5.9 & 3.2 \\ 4.6 & 2.9 \\ 6.2 & 2.8 \\ 4.7 & 3.2 \\ 5.5 & 4.2 \\ 5.0 & 3.0 \\ 4.9 & 3.1 \\ 6.7 & 3.1 \\ 5.1 & 3.8 \\ 6.0 & 3.0 \end{bmatrix}$$

Iteration 1

(assign to nearest of the 3 given centres)

- **red** $\{p_1, p_2, p_4, p_6, p_7, p_9, p_{10}\}$ **green** $\{p_5\}$ **blue** $\{p_3, p_8\}$
- New centres: red $(5.171, 3.171)$, green $(5.5, 4.2)$, blue $(6.45, 2.95)$

Iteration 2

- **red** $\{p_2, p_4, p_6, p_7\}$ **green** $\{p_5, p_9\}$ **blue** $\{p_1, p_3, p_8, p_{10}\}$
- New centres: red $(4.8, 3.05)$, green $(5.3, 4.0)$, blue $(6.2, 3.025)$

Iteration 3

- Every point stays where it was $\rightarrow$ **converged**.

Final answer

Cluster	Members	Final centre
red	p_2, p_4, p_6, p_7	(4.80, 3.05)
green	p_5, p_9	(5.30, 4.00)
blue	p_1, p_3, p_8, p_{10}	(6.20, 3.025)

- **Comment on iterations:** assignments changed at iterations 1 and 2 and were stable at iteration 3, so **convergence took 3 iterations (2 real updates + 1 confirming pass)**.

- Worth adding: the 3 supplied centres were all bunched on the right of the data ($x \approx 6.2$ – 6.6), yet K-means still pulled them apart into a sensible left / middle / right split. Good illustration that it converges quickly even from a poor start.

Problem 1.6 – the defective question 5+3

QUESTION AS PRINTED 2081 Baishakh · Q4 (b) · [5+3]

Use K-means clustering to cluster the following given data for $K = 2$ with Euclidean distance matrix. List down the demerits of this algorithm.

(nothing follows on the paper – the next line is Q5)

The paper prints NO data table for this question – it jumps straight from the sentence to Q5. Verified against the original scan and recorded in the PYQ archive.

In the exam: answer the 3-mark demerits part in full (\$5.2), write the 5 steps of the algorithm, and demonstrate on any small dataset of your own, stating that you assumed one. Use **Problem 1.1's** 6 points – it is the dataset this paper's sister questions use. verified/added

M2. Agglomerative hierarchical clustering

Method

1. Start with each point as its own cluster; write the distance matrix.
 2. Find the **smallest** entry → merge those 2 clusters. **Record the merge height** – that value is the dendrogram bar.
 3. Update the matrix using the linkage rule: **single** = MIN of the two rows, **complete** = MAX.
 4. Repeat until one cluster remains, then draw the dendrogram with merges at their heights.
- **Merge heights never decrease.** If yours does, you have made an arithmetic slip.
 - n points always give exactly $n - 1$ merges.

Problem 2.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q4 (b) · [2+7]

What is hierarchical clustering? Use this clustering approach to draw dendrogram for given data points.

	p_1	p_2	p_3	p_4	p_5
p_1	0.00	0.24	0.22	0.37	0.34
p_2	0.24	0.00	0.15	0.20	0.14
p_3	0.22	0.15	0.00	0.15	0.28
p_4	0.37	0.20	0.15	0.00	0.29
p_5	0.34	0.14	0.28	0.29	0.00
p_6	0.23	0.25	0.11	0.22	0.39

The paper prints the full symmetric 6×6 ; the p_6 column is 0.23, 0.25, 0.11, 0.22, 0.39, 0.00.

Distances are given directly, so no coordinate work. The paper does not name a linkage — use **single link** and say so.

Merge sequence

Step	Merge	Height	Updated distances (MIN)
1	p_3, p_6	0.11	to p_1 0.22, p_2 0.15, p_4 0.15, p_5 0.28
2	p_2, p_5	0.14	to p_1 0.24, p_4 0.20, $\{p_3 p_6\}$ 0.15
3	$\{p_3 p_6\}, p_4$	0.15	to p_1 0.22, $\{p_2 p_5\}$ 0.15
4	$\{p_3 p_4 p_6\}, \{p_2 p_5\}$	0.15	to p_1 0.22
5	all, + p_1	0.22	—

Final answer

Dendrogram heights: **0.11, 0.14, 0.15, 0.15, 0.22.**

- Reading it: $\{p_3, p_6\}$ is the tightest pair; p_1 is the most isolated point, joining last.
- Cutting between 0.15 and 0.22 gives **2 clusters**: $\{p_1\}$ and $\{p_2, p_3, p_4, p_5, p_6\}$.

Problem 2.2 8

QUESTION AS PRINTED 2080 Baishakh · Q7 · [8]

Cluster the following samples based on complete-linkage algorithm and draw the dendrogram. Using Euclidean distance.

Point	x Coordinate	y Coordinate
p1	0.40	0.53
p2	0.22	0.38
p3	0.35	0.32
p4	0.26	0.19
p5	0.08	0.41
p6	0.45	0.30

Step 1: Euclidean distance matrix

$p_1(0.40, 0.53)$ $p_2(0.22, 0.38)$ $p_3(0.35, 0.32)$ $p_4(0.26, 0.19)$ $p_5(0.08, 0.41)$ $p_6(0.45, 0.30)$

	p_1	p_2	p_3	p_4	p_5
p_2	0.234				
p_3	0.216	0.143			
p_4	0.368	0.194	0.158		
p_5	0.342	0.143	0.285	0.284	
p_6	0.235	0.244	0.102	0.219	0.386

Step 2: merge sequence (MAX update)

Step	Merge	Height	Updated distances (MAX)
1	p_3, p_6	0.102	to p_1 0.235, p_2 0.244, p_4 0.219, p_5 0.386
2	p_2, p_5	0.143	to p_1 0.342, p_4 0.284, $\{p_3p_6\}$ 0.386
3	$\{p_3p_6\}, p_4$	0.219	to p_1 0.368, $\{p_2p_5\}$ 0.386
4	$p_1, \{p_2p_5\}$	0.342	to $\{p_3p_4p_6\}$ 0.386
5	the two remaining	0.386	—

Final answer

Heights **0.102, 0.143, 0.219, 0.342, 0.386**; cutting below 0.386 gives $\{p_1, p_2, p_5\}$ and $\{p_3, p_4, p_6\}$.

Problem 2.3 4+4

QUESTION AS PRINTED 2076 Chaitra · Q10 · [4+4]

The table below is a distance matrix for six objects:

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

- Show the final result of hierarchical clustering with single-link by drawing a dendrogram.
- Show the final result of hierarchical clustering with complete-link by drawing a dendrogram.

The ideal question for showing you understand the difference — one matrix, both linkages.

(a) Single link (MIN)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, {CD}	0.16
4	{ABCD}, F	0.20
5	all, + E	0.28

- Heights: **0.12, 0.14, 0.16, 0.20, 0.28**
- Everything merges early and at similar heights → **chaining**.

(b) Complete link (MAX)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, F	0.61
4	{CD}, E	0.70
5	the two groups	0.93

- Heights: **0.12, 0.14, 0.61, 0.70, 0.93**
- Final 2 clusters: **{A,B,F}** and **{C,D,E}** — a balanced split.

The comparison is the marks. The first 2 merges are identical, then the methods diverge completely: single link strings everything onto the A-B-C-D chain and leaves E as a straggler, while complete link produces two clean balanced groups.

Single link chases the nearest neighbour, so it follows chains and elongated shapes. **Complete link** is held back by the worst pair, so it prefers compact globular clusters. verified/added

M3. DBSCAN point classification

Method

1. For each point, count how many points lie within ε (**include the point itself**).
2. Count $\geq \text{MinPts}$ → **core**.
3. Otherwise, within ε of a core point → **border**.
4. Otherwise → **noise**.
5. Join core points within ε of each other into clusters; attach the border points.

Practice

PROBLEM not a PYQ · written to fill the gap verified/added

Classify each of the points $A(1, 1)$, $B(1, 2)$, $C(2, 1)$, $D(8, 8)$, $E(8, 9)$, $F(25, 25)$ as a **core**, **border** or **noise** point under DBSCAN with $\varepsilon = 2$ and $\text{MinPts} = 3$ (Euclidean distance, the point itself counted in its own neighbourhood). Then give the clusters DBSCAN produces.

No PYQ sets a DBSCAN numerical, but the algorithm is asked in 8 papers and a worked classification makes the definitions concrete. verified/added

Step 1: neighbourhood counts

(including the point itself)

Point	Points within $\varepsilon = 2$	Count	Type
$A(1, 1)$	A, B (1.00), C (1.00)	3	core
$B(1, 2)$	B, A (1.00), C (1.41)	3	core
$C(2, 1)$	C, A (1.00), B (1.41)	3	core
$D(8, 8)$	D, E (1.00)	2	border
$E(8, 9)$	E, D (1.00)	2	border
$F(25, 25)$	F only	1	noise

Step 2: form clusters

- A, B, C are all core and all within ε of each other → **Cluster 1** = $\{A, B, C\}$.
- D and E have only 2 neighbours each, and **neither is within ε of any core point** → they are **not border points after all, they are noise**.
- F is isolated → **noise**.

Final answer

- **1 cluster**: $\{A, B, C\}$. **Noise**: D, E, F .
- **The trap**: D and E look like a cluster to the eye, but with $\text{MinPts} = 3$ a pair can never form one. **A border point must be within ε of an actual core point** — being near a non-core point is not enough.
- Lower MinPts to 2 and $\{D, E\}$ becomes a second cluster. **Quote your parameters with the answer.**

Ch 6 – Anomaly / Fraud Detection – Numerical Problems & Solutions

This chapter is **almost entirely theory**. Across all 17 papers there is exactly **one** numerical 76 Ch, on the base rate fallacy. It is solved below, plus 2 practice problems covering the patterns a future numerical would most likely take.

Methods

- **M1** Base rate fallacy: detection rate, false alarm rate, Bayesian detection rate
- **M2** Distance-based outlier: the $DB(r, \pi)$ test

76 Ch
practice

M1. Base rate fallacy

Method

- Let I = anomalous, $\neg I$ = normal, A = alarm raised.
- **Detection rate** = $P(A|I)$ **False alarm rate** = $P(A|\neg I)$
- **Bayesian detection rate** (what an analyst actually cares about):

$$P(I|A) = \frac{P(A|I) P(I)}{P(A|I)P(I) + P(A|\neg I)P(\neg I)}$$

- **Fastest safe route**: invent a round population (10,000), build the confusion matrix, read every rate off it. No formula to misremember.

Problem 1.1 3

QUESTION AS PRINTED 2076 Chaitra · Q11 (b) · [3]

If the probability that a normal object is classified as an anomaly is 0.01 and the probability that an anomalous object is classified as anomalous is 0.99, then what is the false alarm rate and detection rate if 99% of the objects are normal?

Given

- $P(A|\neg I) = 0.01$ (normal object raises an alarm)
- $P(A|I) = 0.99$ (anomalous object raises an alarm)
- $P(\neg I) = 0.99$, so $P(I) = 0.01$

Step 1: the two rates asked for, read straight off the givens

- **False alarm rate** = $P(A|\neg I) = 0.01$ (1%)
- **Detection rate** = $P(A|I) = 0.99$ (99%)

Step 2: why the 99% prior was given – build the matrix on 10,000 objects

	Alarm	No alarm	Total
Anomalous (I)	$0.99 \times 100 = 99$ (TP)	1 (FN)	100
Normal ($\neg I$)	$0.01 \times 9900 = 99$ (FP)	9801 (TN)	9,900
Total	198	9,802	10,000

- Check: detection rate = $99/100 = 0.99$ ✓ false alarm rate = $99/9900 = 0.01$ ✓

Step 3: Bayesian detection rate

$$P(I|A) = \frac{(0.99)(0.01)}{(0.99)(0.01) + (0.01)(0.99)} = \frac{0.0099}{0.0099 + 0.0099} = \frac{99}{198} = 0.50$$

Final answer

- False alarm rate = **0.01**, detection rate = **0.99**.
- **But the Bayesian detection rate is only 0.50** – half of every alarm raised is false, despite a detector that is 99% accurate in both directions.
- **This is the base rate fallacy**. The normal population (9,900) is so much larger than the anomalous one (100) that its 1% error rate manufactures exactly as many false alarms (99) as there are true detections (99).
- **Say this in the exam**: it is the whole point of the question and is worth more than the two arithmetic lines.

If the examiner wanted only the two conditional rates, they are given directly and the question is trivial — which is the clue that the **99% prior** must be used for something. Answer both readings: state the two rates, then compute $P(I|A)$ and name the fallacy. verified/added

Practice A — rarer anomalies

PROBLEM not a PYQ · same detector, rarer anomalies verified/added

The detector of Problem 1.1 is unchanged: a normal object is classified as an anomaly with probability 0.01, and an anomalous object is classified as anomalous with probability 0.99. Now only **1 object in 1,000** is anomalous instead of 1 in 100. Find the detection rate, the false alarm rate, and the probability that an object raising an alarm really is an anomaly.

Shows how fast the Bayesian rate collapses. verified/added

- On 100,000 objects: 100 anomalous, 99,900 normal.
- $TP = 0.99 \times 100 = 99$ $FP = 0.01 \times 99,900 = 999$
- $P(I|A) = \frac{99}{99 + 999} = \frac{99}{1098} = \mathbf{0.090}$
- **Only 9% of alarms are real.** Detection rate and false alarm rate are unchanged at 0.99 and 0.01 — **only the base rate moved.**
- **Conclusion (Axelsson):** to keep $P(I|A)$ usable when $P(I)$ is tiny you must drive the *false alarm rate* down, not push the detection rate up.

M2. Distance-based outlier: the $DB(r, \pi)$ test

Method

- Object o is a $DB(r, \pi)$ -outlier if

$$\frac{|\{o' \mid \text{dist}(o, o') \leq r\}|}{|D|} \leq \pi$$

- In words: count how many objects (including o itself, per Han & Kamber) lie within distance r ; if that count as a fraction of $|D|$ is at most π , it is an outlier.
- Equivalent form: with $k = \lceil \pi |D| \rceil$, o is an outlier if $\text{dist}(o, o_k) > r$.

Practice B

PROBLEM not a PYQ · written to fill the gap verified/added

A dataset contains the points $P_1(1, 1)$, $P_2(1, 2)$, $P_3(2, 1)$, $P_4(2, 2)$ and $P_5(8, 8)$. Taking $r = 2$ and $\pi = 0.4$, and using Euclidean distance, determine which points are $DB(2, 0.4)$ -outliers.

No PYQ has set this yet, but it is the only computable form of the distance-based question that is asked in 6 papers, so it is worth one run-through. verified/added

Step 1: distance matrix

	P_1	P_2	P_3	P_4	P_5
P_1	0	1	1	1.414	9.899
P_2	1	0	1.414	1	9.220
P_3	1	1.414	0	1	9.220
P_4	1.414	1	1	0	8.485
P_5	9.899	9.220	9.220	8.485	0

Step 2: count neighbours within $r = 2$

(threshold $\pi |D| = 0.4 \times 5 = 2$ objects)

- P_1 : itself, $P_2, P_3, P_4 = 4$ $4/5 = 0.8 > 0.4 \rightarrow$ normal
- P_2, P_3, P_4 : 4 each by symmetry $\rightarrow$ normal
- P_5 : only itself = 1 $1/5 = 0.2 \leq 0.4 \rightarrow$ **outlier**

Final answer

P_5 is the only $DB(2, 0.4)$ -outlier.

- **Cross-check with the k -NN form:** $k = \lceil 0.4 \times 5 \rceil = 2$. For P_5 , its 2nd nearest neighbour is at $8.485 > 2 \rightarrow$ outlier $\checkmark$. For P_1 , its 2nd nearest is at $1 \leq 2 \rightarrow$ normal $\checkmark$.

- **Note the sensitivity to r :** raise r to 9 and P_5 stops being an outlier altogether. Quoting the chosen r and π is part of the answer.

Ch 7 – Advanced Applications – Numerical Problems & Solutions

This chapter is almost entirely short notes. Across all 17 papers there is exactly **one** numerical 76 Ch, on PageRank. It is solved in full below, plus 1 practice problem on the variant most likely to be set next.

M1. PageRank

Method

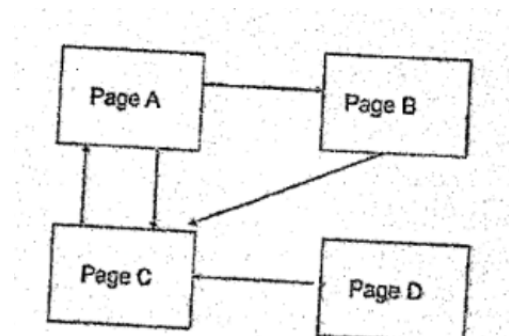
$$PR(A) = (1 - d) + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

1. Write down the link table first: for every page, its **outlinks** (to get C) and its **inlinks** (to know which terms to sum). **Nearly every error comes from confusing the two.**
2. Initialise all ranks (here 0.25).
3. Each iteration, compute all pages using the **previous** iteration's values, then substitute in together.
4. Repeat for the required number of iterations, or until the values settle.
 - A page's own rank is split equally among its outlinks – that is the $/C(T_i)$.
 - A page with **no inlinks** sits permanently at $(1 - d) = 0.15$.

Problem 1.1 8

QUESTION AS PRINTED 2076 Chaitra · Q12 · [8]

Consider the following subset of pages and their links. Apply the PageRank algorithm using a damping factor of 0.85. A minimum of five iterations are required. Assume initial page rank of all pages is 0.25.



Step 1: read the graph

Page	Outlinks	C	Inlinks
A	B, C	2	C
B	C	1	A
C	A	1	A, B, D
D	C	1	none

Step 2: write the 4 equations

($d = 0.85$, $1 - d = 0.15$)

- $PR(A) = 0.15 + 0.85 \frac{PR(C)}{2}$
- $PR(B) = 0.15 + 0.85 \frac{PR(A)}{1}$
- $PR(C) = 0.15 + 0.85 \left(\frac{PR(A)}{2} + PR(B) + PR(D) \right)$
- $PR(D) = 0.15 + 0.85(0) = \mathbf{0.15}$ **constant, D has no inlinks**
- Note A's rank is divided by 2 because A links to both B and C.

Final answer after 5 iterations

- $PR(A) = 0.972$, $PR(B) = 0.519$, $PR(C) = 1.028$, $PR(D) = 0.150$

Step 3: iterate

Iter	PR(A)	PR(B)	PR(C)	PR(D)
0	0.2500	0.2500	0.2500	0.2500
1	0.3625	0.2563	0.6813	0.1500
2	0.7291	0.3041	0.6494	0.1500
3	0.7020	0.4599	0.8459	0.1500
4	0.8690	0.4484	0.9668	0.1500
5	0.9718	0.5193	1.0280	0.1500

Row 0 is the given start value.

Sample working (iteration 1)

- $PR(A) = 0.15 + 0.85(0.25) = 0.3625$
- $PR(B) = 0.15 + 0.85(0.25/2) = 0.2563$
- $PR(C) = 0.15 + 0.85(0.125 + 0.25 + 0.25) = 0.6813$
- $PR(D) = 0.15$

– Ranking: **C > A > B > D**

Two comments worth writing, they show understanding. verified/added

1. The values are still climbing at iteration 5 but the **ranking has been stable since iteration 3**. PageRank converges on *order* much faster than on *value*, which is why 5 iterations is enough for the question.
2. Solving the equations exactly gives the fixed point $PR(A) = 1.490$, $PR(B) = 0.783$, $PR(C) = 1.577$, $PR(D) = 0.150$, and these sum to **4.00 = N**. **That sum is your check** – in this (non-normalised) form the ranks always converge to the number of pages.
3. **C wins** because it collects links from three pages while A collects only one, and **D is stuck at 0.15** because nothing points to it. Linking out never raises your own rank.

Practice – the normalised form

PROBLEM not a PYQ · the same graph, normalised form verified/added

Repeat Problem 1.1 on the same four-page graph, but using the *normalised* PageRank formula

$$PR(A) = \frac{1-d}{N} + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

with $d = 0.85$, $N = 4$ and an initial rank of 0.25 for every page. Compare the converged ranks and the ordering with those of Problem 1.1.

This form makes the ranks sum to 1 instead of N . Know both. verified/added

- With $N = 4$, $\frac{1-d}{N} = \frac{0.15}{4} = 0.0375$.
- Iteration 1 from the same 0.25 start:
 - $PR(A) = 0.0375 + 0.85(0.25) = 0.2500$
 - $PR(B) = 0.0375 + 0.85(0.125) = 0.1438$
 - $PR(C) = 0.0375 + 0.85(0.625) = 0.5688$
 - $PR(D) = 0.0375$
- Sum = 1.0001 ≈ 1 ✓ – **that is the check for this form.**
- **The ranking is identical** ($C > A > B > D$); only the scale differs.
- **In the exam:** state which form you are using in one line. Either is accepted; mixing them is not.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.